ANALYSIS / POLYLOGARITHMS / TRANSSERIES

Polylogarithm Order Jets and Resonant Inversion

Convergent logarithmic charts, complex branch splitting, and exact reconstruction of exponential sectors

A research report prepared for Vladimir Reshetnikov

Prepared with OpenAI ChatGPT

12 October 2026 • Version 1.0

Scope. Complete proofs of explicit extensions of the ProveIt polylogarithm and transseries programmes, together with symbolic and high-precision numerical checks. Classical formulas and predecessor theorems are attributed separately. No claim of established literature priority or resolution of an unrelated special-value conjecture is made.

Repository snapshot: `9ec0f0406afe7a5475b03ff65ed91f7e283979b8`.
The accompanying package contains the self-contained LaTeX source, this PDF, verification programs, recorded results, and integration notes.

Abstract

Differentiation in the order of a polylogarithm changes the asymptotic information visible at an integer order. Near the singular argument it raises logarithmic degree; at large negative argument it restores factorially divergent tails to otherwise terminating Sommerfeld expansions. We develop both phenomena in explicit analytic coordinates. A reciprocal-Gamma change of finitely many order derivatives reduces every Taylor-subtracted integer-order Lerch jet to $t^p\{\log^q(1/t)+tR(t)\}$ with R holomorphic. Its inverse is an actually convergent function of two Lambert-W coordinates. We give finite Lagrange formulas for every inverse coefficient, prove the sharp pole order $2n-1$, and identify its Catalan leading constant. For the direct inverse of $\zeta(s)-\mathrm{Li}_s(e^{-t})$ near $s=2$, we construct a convergent chart for an exponential lattice of critical points and critical values, and prove local exhaustion on the principal sheet for small nonzero imaginary order displacement. The nonzero imaginary zeros of the leading crossover profile produce two local inverse branches separated on a square-root inverse-logarithmic scale. At infinity an explicit Borel transform yields exact lateral sums and shows that a single order derivative recovers the entire exponentially small polylogarithm at the opposite argument. Its divergent tail has a meromorphic Borel kernel obtained from $\pi\xi/\sin(\pi\xi)$; every exponential coefficient is a residue. We also prove the fixed-sector complex extension requested in a specific repository research question. All branch conventions, convergence statements, and limits of the numerical evidence are stated explicitly.

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1 Questions, provenance, and the main results

The two repository programmes meet at a natural question: what is lost when a parameter-dependent special-function expansion is specialized at an integer before it is inverted or differentiated? There are two particularly clean answers. Near $z = 1$, colliding Gamma and zeta poles leave logarithms, and order derivatives leave powers of logarithms. Near $z = -\infty$, reciprocal-Gamma zeros terminate a power expansion, although exponentially small terms remain. These are distinct analytic regimes, but the order parameter gives a concrete way to study both.

The purpose of this report is to prove several explicit results at this intersection, not to identify numerical fits with proofs. The principal objects are ordinary polylogarithms, their derivatives with respect to the order, and the shifted family

$$F(s,t,a) = e^{-at}\Phi(e^{-t},s,a) = \sum_{n=0}^{\infty} \frac{e^{-(n+a)t}}{(n+a)^s}, \qquad \Re t > 0, \quad \Re a > 0. \tag{1.1}$$

Here the powers of $n+a$ use the logarithm in the right half-plane. For $a = 1$, $F(s,t,1) = \mathrm{Li}_s(e^{-t})$. The continuation in t used below is specified by the paired expansion in section 2.

1.1 What the existing material already proves

The inspected snapshot is `9ec0f0406afe7a5475b03ff65ed91f7e283979b8`. The polylogarithm programme has a collective manuscript and many preserved predecessor reports. Its resonance continuation already proves a forward expansion uniform in the nonnegative integer order, the distance of the order from that integer, and a compact set of positive shifts [1]. It also treats ordinary integer-order derivatives and their Stieltjes finite parts. We use and attribute that work.

The transseries programme already proves a convergent Lambert chart for $Bx^m(\log(1/x) + h) + x^{m+1}R(x)$, as well as a confluent chart for a nearby nonintegral exponent [2, 3]. Consequently, neither elementary pole cancellation nor the existence of a single-logarithm Lambert chart would be a sufficient contribution. The extensions developed here concern higher logarithmic degree, explicit inverse coefficient structure, and complex degeneracies of the leading crossover profile.

The classical literature must also be separated from the present derivations. Jonquière and Lerch expansions are standard [4, 5]; numerical pairing near integer order is explicit in Roughan's work [8]. Generalized polylogarithms with logarithmic weights appear in analytic combinatorics [10], and integer-order derivative calculations appear in Bailey and Borwein [9]. Complete Fermi–Dirac asymptotics, including exponentially small terms, were studied by Garoni, Frankel, and Glasser and are discussed by Gil, Segura, and Temme [16, 17]. Fractional polylogarithm resurgence has a separate established literature [14]. The zero geometry also has a substantial literature; in particular, O'Sullivan surveys general polylogarithm zeros while proving global dilogarithm results [19]. Our local critical-value and inverse-fold arguments have a different scope.

1.2 The increments proved here

Result	Precise increment over the inputs above
Complex forward resonance	The fixed-sheet, closed-sector version of the predecessor's question R1, still uniform in the integer order.
Gamma-normalized jets	A finite triangular transformation converts arbitrary order jets to one pure logarithmic monomial plus an analytic tail, with explicit Hurwitz-zeta derivative coefficients.
Inverse coefficient theorem	A convergent two-coordinate chart for every $p, q \geq 1$, an all-order coefficient formula, and sharp Catalan pole constants.
Complex inverse resonance	A common chart classifies the small imaginary-order critical lattice; at each nonzero $2\pi i k$, two inverse branches split on an inverse-square-root-log scale; a transverse unfolding is explicit.
Borel reconstruction	Exact lateral identities and a meromorphic first-order-jet kernel whose residues recover the complete opposite- argument exponential series.

The table records the mathematical scope of this report. The statements are proved below, but the literature search is not a proof of global priority. The complex branch-splitting theorem and the explicit combination of Gamma normalization, inverse coefficient poles, and order-jet Borel reconstruction are the strongest candidates for further originality review.

1.3 Conventions and types of conclusion

We use $\zeta^{[r]}(s, a)$ for differentiation in the first argument, and

$$\zeta(1+\varepsilon, a) = \frac{1}{\varepsilon} + \sum_{r=0}^{\infty} \frac{(-1)^r \gamma_r(a)}{r!} \varepsilon^r. \tag{1.2}$$

Thus $\gamma_0(1) = \gamma$, $\gamma_0(a) = -\psi(a)$, and $\gamma_1 = \gamma_1(1)$. An order jet means one of these derivatives or a derivative of Li_s with respect to s, not with respect to its argument. All displayed coefficient extractions are ordinary Taylor coefficients; a derivative of order r is $r!$ times the coefficient.

An exact analytic chart means equality after evaluation of a convergent multivariable power series at its specified small coordinates. Expanding its Lambert core further gives a power–logarithmic transseries. The Borel statements at infinity use a different, explicitly stated summation convention. Analytic continuation around $t = 0$ and lateral Borel summation across a singular ray are not interchangeable operations.

The verification programs check finite symbolic consequences and numerical examples. They do not establish analytic continuation, uniformity in unbounded parameters, existence of a branch, or absence of additional global branches. Those claims require the proofs.

2 Paired Lerch expansions on complex sectors

The inspected repository revision is `9ec0f0406afe7a5475b03ff65ed91f7e283979b8`. The direct predecessor is the resonance continuation [1], whose Theorem `res:main` is stated for positive real t. Its accompanying research question R1 under "Uniform resonance and Stieltjes finite parts", asks for complex-sector continuation with a specified logarithm and a majorant uniform on closed subsectors. The result below answers that precise question. It does not assert global continuation across the neighboring singularities $\pm 2\pi i$.

Write

$$F(s,t,a) = e^{-at}\Phi(e^{-t},s,a),$$
$$\mathcal{R}_k(\epsilon,t,a) = \frac{k!}{(-t)^k}\left[F(k+1+\epsilon,t,a) - \sum_{j=0}^{k-1}\zeta(k+1+\epsilon-j,a)\frac{(-t)^j}{j!}\right].$$

For complex t, F denotes the continuation from $t>0$ to the chosen slit disk. The initial series defining F is needed only where $\operatorname{Re} t > 0$; the continued function is fixed by the exact Lerch expansion.

Let $k \geq 0$, let $H_k^{(r)} = \sum_{j=1}^{k} j^{-r}$, and put

$$T = -\operatorname{Log} t + H_k - \gamma, \qquad E_k(w) = \exp\left(\sum_{r=2}^{\infty}\frac{\zeta(r)+(-1)^r H_k^{(r)}}{r}w^r\right) = \sum_{j\geq 0} b_{k,j}w^j.$$

In particular $b_{k,0}=1$, $b_{k,1}=0$. Use the standard convention

$$\zeta(1+w,a) = \frac{1}{w} + \sum_{r\geq 0}\frac{(-1)^r\gamma_r(a)}{r!}w^r.$$

Theorem 2.1 (Uniform resonance on complex sectors)**.** *Fix $0<t_0<2\pi$, $0<\delta<\pi$, a compact set $A\subset\{a:\operatorname{Re} a>0\}$, a compact set $U\subset\mathbb{C}$, and nonnegative integers M,J,D. Give t its principal logarithm and restrict it to*

$$0<|t|\leq t_0, \qquad |\arg t| \leq \pi-\delta.$$

As $\operatorname{Re} T\to+\infty$, uniformly in these choices and in all integers $k\geq 0$,

$$\mathcal{R}_k(u/T,t,a) = Th(u) + \gamma_0(a) + \sum_{r=1}^{M}\frac{u^r}{T^r}\left(\frac{(-1)^r\gamma_r(a)}{r!} - e^{-u}b_{k,r+1}\right) + O(|T|^{-M-1}),$$
$$h(u) = \frac{1-e^{-u}}{u}, \qquad h(0)=1.$$

The same bound holds after at most J derivatives in u and at most D complex derivatives in a. The analytic remainder separated before the power expansion satisfies the sharper estimate

$$B_k(u/T,t,a) = O\left(\frac{|t|}{k+1}\right) = O(e^{-\operatorname{Re} T}).$$

The latter estimate is also valid with the stated fixed derivatives.

Proof. On the slit disk the classical Lerch expansion [5] continues as

$$F(s,t,a) = \Gamma(1-s)t^{s-1} + \sum_{j\geq 0}\zeta(s-j,a)\frac{(-t)^j}{j!}, \qquad |t|<2\pi,$$

where $t^{s-1} = \exp((s-1)\operatorname{Log} t)$. Pole pairing at $s=k+1+\epsilon$ gives the exact identity

$$\mathcal{R}_k(\epsilon,t,a) = \frac{1-e^{-T\epsilon}E_k(\epsilon)}{\epsilon} + G_a(\epsilon) + B_k(\epsilon,t,a),$$
$$G_a(\epsilon) = \zeta(1+\epsilon,a) - \frac{1}{\epsilon},$$
$$B_k(\epsilon,t,a) = \sum_{\ell\geq 1}\frac{k!(-t)^\ell}{(k+\ell)!}\zeta(1+\epsilon-\ell,a).$$

Every apparent singularity at $\epsilon = 0$ is removable. This identity first follows for positive t and then on the slit disk by analytic continuation. It also follows directly from the gamma recurrence with the chosen logarithm.

Choose $0 < \eta < 1/2$ and $t_0 < R < 2\pi$. The functions E_k are uniformly bounded on $|w| \le \eta$, because

$$\left|\zeta(r) + (-1)^r H_k^{(r)}\right| \le 2\zeta(r).$$

Consequently all finite Taylor remainders on smaller disks have bounds independent of k. The same is true for G_a when a ranges in a slightly enlarged compact subset of $\operatorname{Re} a > 0$.

Here are details for the common tail majorant. Hermite's Hurwitz formula [6] in its difference-of-powers form reads

$$\zeta(s,a) = \frac{a^{-s}}{2} + \frac{a^{1-s}}{s-1} + i\int_0^\infty \frac{(a+iy)^{-s} - (a-iy)^{-s}}{e^{2\pi y} - 1}\, dy.$$

All powers use principal logarithms; $\operatorname{Re}(a \pm iy) > 0$. For $s = 1 + \epsilon - \ell$, the two elementary terms have factorially summable majorants, uniformly for $|\epsilon| \le \eta$ and the chosen compact a set. For $y \ge 1$, summing absolute majorants weighted by $R^\ell/\ell!$ leaves a fixed polynomial in y times $e^{-(2\pi - R)y}$. For $0 \le y \le 1$, subtracting the two powers first, or integrating their derivative along the segment between $a - iy$ and $a + iy$, gains a factor y. This cancels the first-order zero of the denominator, and the remaining factorial series is uniformly bounded. Thus

$$\sup_{a, |\epsilon| \le \eta} \sum_{\ell \ge 1} \frac{t_0^{\ell-1}}{\ell!} |\zeta(1 + \epsilon - \ell, a)| < \infty.$$

Using

$$\frac{k!}{(k+\ell)!} \le \frac{1}{(k+1)\ell!} \quad (\ell \ge 1)$$

now gives $|B_k| \le C|t|/(k+1)$. Finally,

$$\frac{|t|}{k+1} = e^{-\operatorname{Re} T} \frac{e^{H_k - \gamma}}{k+1} \le e^{-\operatorname{Re} T},$$

since $H_k - \gamma \le \log(k+1)$.

Put $\epsilon = u/T$. On the stated sector $|\operatorname{Im} T| \le \pi - \delta$, so $|T| \sim \operatorname{Re} T$ uniformly as the latter tends to infinity. The paired term becomes

$$Th(u) - e^{-u} \sum_{j \ge 2} b_{k,j} (u/T)^{j-1}.$$

Its Taylor expansion together with that of $G_a(u/T)$ yields the asserted remainder $O(|T|^{-M-1})$. The analytic tail is smaller than every fixed power of $|T|^{-1}$. To obtain the derivative estimates, prove the same bounds first on enlarged compact sets in u and a, then apply Cauchy's formula. All constants remain independent of k, t. □

Limits of the statement. The theorem treats a fixed logarithmic sheet. It does not assert the same algebraic remainder when winding number grows without restriction: then $|\operatorname{Im} T|$ need not be bounded and an error controlled by $e^{-\operatorname{Re} T}$ need not be smaller than $|T|^{-N}$. A logarithmic-cover version follows under the explicit additional condition $\operatorname{Re} T \ge c|T| \to \infty$ for fixed $c > 0$. No singularity of the Lerch series outside the radius 2π is crossed in either version.

3 From order derivatives to pure logarithmic normal forms

In this section $p \geq 1$ and $q \geq 1$ are fixed integers. The restriction $p \geq 1$ makes the normal form vanish at $t = 0$ and hence gives a small inverse. The coefficient identities themselves also make sense for $p = 0$. Let

$$\mathcal{T}_p(\varepsilon, t, a) = F(p + 1 + \varepsilon, t, a) - \sum_{j=0}^{p-1} \zeta(p + 1 + \varepsilon - j, a) \frac{(-t)^j}{j!}, \tag{3.1}$$

$$\mathcal{R}_p(\varepsilon, t, a) = \frac{p!}{(-t)^p} \mathcal{T}_p(\varepsilon, t, a), \qquad L = -\operatorname{Log} t. \tag{3.2}$$

The polynomial subtraction in (3.1) is part of the definition. For $p > 1$, the unsubtracted difference $\zeta(p + 1) - \operatorname{Li}_{p+1}(e^{-t})$ has a nonzero linear leading term. Its inverse is asymptotically linear with logarithmic corrections, and is not holomorphic at the endpoint. The subtraction makes the resonant term of order t^p the leading scale.

3.1 The raw logarithmic polynomials

Set $H_p = \sum_{j=1}^{p} j^{-1}$ and

$$A_p(\varepsilon) = \frac{\Gamma(1 - \varepsilon)}{\prod_{j=1}^{p}(1 + \varepsilon/j)}, \qquad D_p(\varepsilon) = A_p(\varepsilon)^{-1}, \tag{3.3}$$

$$X = L + H_p - \gamma, \qquad E_p(\varepsilon) = e^{(H_p - \gamma)\varepsilon} A_p(\varepsilon) = \sum_{j \geq 0} b_{p,j} \varepsilon^j. \tag{3.4}$$

In particular $b_{p,0} = 1$, $b_{p,1} = 0$, and

$$\log E_p(\varepsilon) = \sum_{j \geq 2} \frac{\zeta(j) + (-1)^j H_p^{(j)}}{j} \varepsilon^j, \qquad H_p^{(j)} = \sum_{h=1}^{p} h^{-j}. \tag{3.5}$$

The paired expansion gives

$$\mathcal{R}_p(\varepsilon, t, a) = \frac{1 - e^{-X\varepsilon} E_p(\varepsilon)}{\varepsilon} + G_a(\varepsilon) + B_p(\varepsilon, t, a), \tag{3.6}$$

where $G_a = \zeta(1 + \varepsilon, a) - 1/\varepsilon$ and

$$B_p(\varepsilon, t, a) = \sum_{\ell \geq 1} \frac{p!(-t)^\ell}{(p + \ell)!} \zeta(1 + \varepsilon - \ell, a). \tag{3.7}$$

Its analytic tail is holomorphic in t for $|t| < 2\pi$, locally in ε and a in the stated domains.

Proposition 3.1 (Raw jet polynomial)**.** *For every $r \geq 0$,*

$$(-1)^r \partial_\varepsilon^r \mathcal{R}_p(0, t, a) = P_{p,r,a}(X) + (-1)^r \partial_\varepsilon^r B_p(0, t, a), \tag{3.8}$$

where

$$P_{p,r,a}(X) = \gamma_r(a) + (-1)^{r+1} r! \sum_{j=0}^{r+1} b_{p,j} \frac{(-X)^{r+1-j}}{(r + 1 - j)!}. \tag{3.9}$$

This is a degree-$r + 1$ polynomial with leading coefficient $1/(r + 1)$. The first three are

$$\begin{aligned} P_{p,0,a}(X) &= X + \gamma_0(a), \\ P_{p,1,a}(X) &= \tfrac{1}{2} X^2 + b_{p,2} + \gamma_1(a), \\ P_{p,2,a}(X) &= \tfrac{1}{3} X^3 + 2b_{p,2} X - 2b_{p,3} + \gamma_2(a). \end{aligned} \tag{3.10}$$

Proof. Extract $(-1)^r r![\varepsilon^r]$ in (3.6). Normal convergence permits extraction inside (3.7). The term $b_{p,0}$ supplies the leading coefficient. □

The polynomial has a simple origin, but replacing it by its leading term before inversion discards an infinite family of inverse-logarithmic corrections. We next construct a finite change of coordinates that removes all its lower powers exactly.

3.2 A finite reciprocal-Gamma transformation

Definition 3.2 (Gamma-normalized order jet)**.** Define the germ holomorphic in ε at zero

$$\mathcal{K}_p(\varepsilon,t,a)=\varepsilon D_p(\varepsilon)\mathcal{R}_p(\varepsilon,t,a)-D_p(\varepsilon)\{\varepsilon\zeta(1+\varepsilon,a)\}, \tag{3.11}$$

and set

$$\mathcal{Y}_{p,q,a}(t)=(-1)^{q+1}t^p\partial_\varepsilon^q\mathcal{K}_p(0,t,a). \tag{3.12}$$

The braces in (3.11) emphasize the removable product; neither factor containing a pole is evaluated separately at zero.

Theorem 3.3 (Pure logarithmic normal form)**.** *For fixed $p,q\ge 1$ and $\Re a>0$,*

$$\boxed{\mathcal{Y}_{p,q,a}(t)=t^p\{L^q+tR_{p,q,a}(t)\}},\qquad L=-\operatorname{Log}t, \tag{3.13}$$

where $R_{p,q,a}$ is single-valued and holomorphic for $|t|<2\pi$. Its complete coefficient formula is

$$[t^j]R_{p,q,a}(t)=\frac{(-1)^{q+j}p!q}{(p+j+1)!}\,\partial_\varepsilon^{q-1}\{D_p(\varepsilon)\zeta(\varepsilon-j,a)\}\Big|_{\varepsilon=0}\quad(j\ge 0). \tag{3.14}$$

Only order derivatives of F and its Taylor subtraction through order $q-1$ occur in $\mathcal{Y}$. The transformation is triangular with a nonzero diagonal, and its remaining constants are finite expressions in generalized Stieltjes constants and derivatives of D_p at zero.

Proof. Before centering the Gamma factor, the paired identity reads

$$\mathcal{R}_p=\zeta(1+\varepsilon,a)-A_p(\varepsilon)t^\varepsilon/\varepsilon+B_p.$$

Multiplication by εD_p and the indicated subtraction give the exact identity

$$\mathcal{K}_p(\varepsilon,t,a)=-t^\varepsilon+\varepsilon D_p(\varepsilon)B_p(\varepsilon,t,a). \tag{3.15}$$

Since $t^\varepsilon=e^{-L\varepsilon}$, applying (3.12) to the first term gives t^pL^q. For the second term, $\partial_\varepsilon^q(\varepsilon H)(0)=q\partial_\varepsilon^{q-1}H(0)$. Substitute (3.7), put $\ell=j+1$, and collect the signs. This proves (3.14) and convergence.

For an entirely finite expression, put $d_j=[\varepsilon^j]D_p(\varepsilon)$ and $c_q(a)=[\varepsilon^q]\{\varepsilon D_p(\varepsilon)\zeta(1+\varepsilon,a)\}$. Then

$$\mathcal{Y}_{p,q,a}(t)=(-1)^{q+1}q!\left\{(-1)^p p!\sum_{j=0}^{q-1}d_j[\varepsilon^{q-1-j}]\mathcal{T}_p(\varepsilon,t,a)-c_q(a)t^p\right\}. \tag{3.16}$$

Its coefficient of the highest derivative $\partial_\varepsilon^{q-1}\mathcal{T}_p(0,t,a)$ is $(-1)^{p+q+1}p!q\neq 0$. Thus no infinite combination of jets, or unknown asymptotic subtraction, is hidden in the definition. □

The coefficient ring is explicit. From (3.5),

$$\begin{aligned}D_p'(0)&=H_p-\gamma,\\ D_p''(0)&=(H_p-\gamma)^2-\zeta(2)-H_p^{(2)},\end{aligned} \tag{3.17}$$

and all higher derivatives follow by Bell-polynomial expansion of

$$\log D_p(\varepsilon) = (H_p - \gamma)\varepsilon - \sum_{j\geq 2} \frac{\zeta(j) + (-1)^j H_p^{(j)}}{j} \varepsilon^j.$$

Together with (1.2), this is a finite algorithm for every normalizing constant and every tail coefficient.

Example 3.4 (Two explicit dilogarithm jets)**.** For $p = 1, a = 1$ the first normalized jet is

$$\mathcal{Y}_{1,1,1}(t) = \zeta(2) - \mathrm{Li}_2(e^{-t}) - t = tL + \frac{t^2}{4} - \frac{t^3}{72} + O(t^5). \tag{3.18}$$

For $q = 2$, let

$$c_2 = -\tfrac{1}{2}(\gamma^2 + \zeta(2)) - \gamma_1.$$

Writing $\mathrm{Li}_2^{[1]}(z) = \partial_s \mathrm{Li}_s(z)|_{s=2}$, the finite identity is

$$\mathcal{Y}_{1,2,1}(t) = 2\{ \mathrm{Li}_2^{[1]}(e^{-t}) - \zeta'(2) + (1-\gamma)(\mathrm{Li}_2(e^{-t}) - \zeta(2)) + c_2 t\}. \tag{3.19}$$

It has the convergent local expansion

$$\mathcal{Y}_{1,2,1}(t) = tL^2 - \frac{1 - \gamma + \log(2\pi)}{2}\, t^2 + O(t^3). \tag{3.20}$$

The coefficient follows from $\zeta(0) = -1/2$ and $\zeta'(0) = -\frac{1}{2}\log(2\pi)$.

Example 3.5 (A shift removes the first analytic correction)**.** For $q = 1, a = 1/2$, $\zeta(0, 1/2) = 0$ and, more generally, $\zeta(-2j, 1/2) = 0$ for every $j \geq 0$. Therefore $B_p(0, t, 1/2)$ has only even powers of t and

$$\mathcal{Y}_{p,1,1/2}(t) = t^p \left\{ L + \frac{t^2}{24(p+1)(p+2)} + O(t^4) \right\}. \tag{3.21}$$

This follows from $\zeta(s, 1/2) = (2^s - 1)\zeta(s)$, including its continued value at $s = 0$. At $q \geq 2$, derivatives of these zeros generally restore odd tail powers. This is a local counterpart of the integer-order degeneration studied at infinity below.

3.3 Exact local monodromy

Proposition 3.6 (Logarithmic degree is detected by monodromy)**.** *Let M denote one positive continuation around $t = 0$. Then*

$$(M - 1)\mathcal{Y}_{p,q,a}(t) = t^p\{(L - 2\pi\mathrm{i})^q - L^q\}, \tag{3.22}$$

and

$$(M - 1)^q \mathcal{Y}_{p,q,a} = (-2\pi\mathrm{i})^q q! t^p \neq 0, \qquad (M - 1)^{q+1}\mathcal{Y}_{p,q,a} = 0. \tag{3.23}$$

In particular no finite ramified change $t = w^N$ makes the function single-valued near zero.

Proof. The analytic tail in (3.13) is single-valued, t^p is unchanged, and L changes to $L - 2\pi\mathrm{i}$. Repeated finite differences of a degree-q polynomial give (3.23). A nonzero integer multiple of $2\pi\mathrm{i}$ cannot be a period of L^q. □

4 Convergent Lambert charts and exact inverse coefficients

We first prove a statement for an arbitrary analytic tail. Its holomorphic conclusions then apply to every normalized polylogarithm jet in theorem 3.3. This isolates the analytic inversion from the arithmetic expressions for the coefficients.

Theorem 4.1 (Convergent inverse for arbitrary logarithmic degree)**.** *Let $p, q \geq 1$ be integers and let R be holomorphic near zero and real on the real axis. For sufficiently small $y > 0$, the equation*

$$y = t^p\{\log^q(1/t) + tR(t)\} \tag{4.1}$$

has a unique sufficiently small positive solution. Define

$$\ell = -\frac{q}{p}W_{-1}\left(-\frac{p}{q}y^{1/q}\right), \qquad T = e^{-\ell}, \qquad \eta = \ell^{-1}. \tag{4.2}$$

There is a holomorphic function $V(T, \eta)$ near $(0, 0)$, with $V(0, 0) = 1$, such that the solution is exactly

$$t = TV(T, \eta), \tag{4.3}$$

and V is uniquely determined near 1 by

$$V^p(1 - \eta \log V)^q + T\eta^q V^{p+1}R(TV) = 1. \tag{4.4}$$

Moreover,

$$V - 1 \in T\eta^q\mathbb{C}\{T, \eta\}. \tag{4.5}$$

If $R(t) = t^h\widetilde{R}(t)$, $h \geq 0$, then the stronger inclusion $V - 1 \in T^{h+1}\eta^q\mathbb{C}\{T, \eta\}$ holds. Compatible branches extend the identity to a sufficiently small sector about the positive y-axis.

Proof. The defining identity for W_{-1} yields $\ell e^{-p\ell/q} = y^{1/q}$, hence

$$y = T^p\ell^q, \qquad \log(1/T) = \ell. \tag{4.6}$$

Substituting $t = TV$ in (4.1) and dividing by (4.6) gives (4.4). Its left side minus 1 is holomorphic near $(T, \eta, V) = (0, 0, 1)$, where the derivative in V is $p \neq 0$. The analytic implicit function theorem therefore supplies the convergent germ.

To obtain the stronger support, factor

$$V^p(1 - \eta \log V)^q - 1 = (V - 1)J(V, \eta), \qquad J(1, 0) = p.$$

The function J is a unit in the local ring. Equation (4.4) expresses $V - 1$ as $T\eta^q$ times a holomorphic function, and as $T^{h+1}\eta^q$ times one if R has the asserted zero.

For the real branch, $\ell \to \infty$, so $(T, \eta) \to (0, 0)$ and $V \to 1$. The derivative of the right side of (4.1) is

$$t^{p-1}L^{q-1}(pL - q) + O(t^p), \qquad L = \log(1/t),$$

which is positive for small $t > 0$. This proves uniqueness of the small real solution. The holomorphic germ and the chosen Lambert branches give the stated sectorial continuation. □

The proof uses a standard implicit-function mechanism, already present at $q = 1$ in [2]. The higher-degree support $T\eta^q$, the exact special-function reduction, and the coefficient structure proved next are the additional information.

For a complex-valued analytic tail, the same implicit-function proof gives the holomorphic chart and all coefficient statements. The positive-real solution assertion uses the stated reality assumption. Thus for the Lerch applications it applies when $a > 0$; a complex shift uses the holomorphic chart with compatible branches.

4.1 All coefficients by a finite Lagrange formula

Write $R(t) = \sum_{j\geq 0} r_j t^j$ and

$$V(T,\eta) = 1 + \sum_{n\geq 1} v_n(\eta) T^n. \tag{4.7}$$

Put

$$\begin{aligned} D &= p - q\eta, \\ h(z,\eta) &= \frac{(1+z)^p(1-\eta\log(1+z))^q - 1}{z}, \qquad h(0,\eta) = D. \end{aligned} \tag{4.8}$$

The quotient is interpreted by continuity at $z = 0$.

Theorem 4.2 (Finite inverse coefficient formula)**.** *For every $n \geq 1$,*

$$\boxed{v_n(\eta) = \sum_{k=1}^{n} \frac{(-\eta^q)^k}{k} [t^{n-k}] R(t)^k \, [z^{k-1}] \frac{(1+z)^{kp+n}}{h(z,\eta)^k}.} \tag{4.9}$$

Each coefficient is a rational function of η, divisible by η^q at zero. Its only possible finite pole is $\eta = p/q$, of order at most $2n-1$. The first two coefficients are

$$v_1 = -\frac{\eta^q r_0}{D}, \tag{4.10}$$

$$v_2 = -\frac{\eta^q r_1}{D} + \frac{\eta^{2q} r_0^2}{D^3} \left\{ (p+1)D - \frac{D^2 - D - q\eta^2}{2} \right\}. \tag{4.11}$$

Proof. Set $w = V - 1$. Equation (4.4) becomes

$$w = -\frac{\eta^q}{h(w,\eta)} \sum_{j\geq 1} r_{j-1} T^j (1+w)^{p+j}.$$

Introduce an auxiliary multiplier on the right and apply the classical Lagrange inversion formula [11]. At fixed degree T^n, only powers $k \leq n$ of that multiplier can contribute, so setting it equal to 1 is legitimate both formally and for the convergent germ. When the k factors carry indices $j_1 + \cdots + j_k = n$, their power of $(1+w)$ is $kp + n$, independent of the composition. Their coefficient is $[t^{n-k}]R(t)^k$. This gives (4.9).

At each fixed degree in z, the Taylor coefficients of h are polynomials in η. To extract z^{k-1} in h^{-k} requires at most $k-1$ additional factors of D^{-1} beyond its initial D^{-k}. Thus the denominator has power at most $2k - 1 \leq 2n - 1$. The factor η^{qk} gives the stated divisibility. Finally,

$$\frac{\partial^2}{\partial V^2} \{V^p (1 - \eta \log V)^q\} \Big|_{V=1} = D^2 - D - q\eta^2.$$

Expanding through T^2 proves (4.10)–(4.11). □

Theorem 4.3 (Sharp poles and their Catalan constants)**.** *Suppose $r_0 \neq 0$, set $\eta_0 = p/q$, and write $C_n = (n+1)^{-1}\binom{2n}{n}$. Then every v_n has a pole of exact order $2n-1$ at η_0, with*

$$\lim_{\eta\to\eta_0} (p - q\eta)^{2n-1} v_n(\eta) = -C_{n-1} (\eta_0^q r_0)^n \left(-\frac{p^2}{2q} \right)^{n-1}. \tag{4.12}$$

Proof. In (4.9), only $k = n$ can have the maximum denominator power. Write $h(z, \eta) = D + h_1(\eta)z + \cdots$. The maximum-pole contribution to $[z^{n-1}]h^{-n}$ is

$$(-1)^{n-1}\binom{2n-2}{n-1}h_1(\eta)^{n-1}D^{-(2n-1)}.$$

The numerator has constant term 1 and any positive-degree contribution lowers the maximum pole order. Since $h_1(\eta_0) = -p^2/(2q)$, multiplication by $(-\eta^q)^n r_0^n/n$ gives (4.12). All its factors are nonzero. □

These poles describe the coefficient functions as rational functions of the auxiliary coordinate. They do not assert that the actual inverse, with its coordinates constrained by (4.6), has its nearest singularity at that point. The value $\eta = p/q$ corresponds to the critical point of the unperturbed logarithmic core; it lies outside the asymptotic regime $\eta \to 0$.

4.2 Convergence, transseries support, and examples

Choose a closed bidisk $|T| \le r$, $|\eta| \le s$ inside the domain of the chart and a bound $|V| \le C$ there. Cauchy's estimates imply the explicit geometric truncation inequality

$$\left| t - T\sum_{j=0}^{J}\sum_{k=0}^{K}[T^j\eta^k]V\,T^j\eta^k \right| \le |T|C\,\frac{(|T|/r)^{J+1} + (|\eta|/s)^{K+1}}{(1-|T|/r)(1-|\eta|/s)}. \tag{4.13}$$

This asserts existence of such constants, not a computed numerical bidisk. The supplied numerical checks are not interval certificates for r, s, C.

Along the positive axis,

$$t \sim p^{q/p}\,y^{1/p}\{\log(1/y)\}^{-q/p}. \tag{4.14}$$

Indeed $p\ell - q\log\ell = \log(1/y)$, so $\ell \sim \log(1/y)/p$, and $V \to 1$. The logarithmic factor proves that no nonzero convergent Puiseux series in y represents the small inverse. On the other hand, the exact Lambert coordinates give a convergent representation and an explicit all-order recurrence. Expanding W_{-1} produces the familiar nested logarithms; convergence questions for those expansions have been studied independently [12, 13].

For (3.20), take $p = 1, q = 2$ and $r_0 = -(1 - \gamma + \log(2\pi))/2$. The leading correction is

$$t = T\left\{1 + \frac{1-\gamma+\log(2\pi)}{2}\,\frac{T\eta^2}{1-2\eta} + O(T^2)\right\}, \tag{4.15}$$

uniformly for η in a sufficiently small closed disk. For the half-shift example (3.21), the absence of the first tail term strengthens the chart support to $V - 1 \in T^2\eta\mathbb{C}\{T, \eta\}$. Thus arithmetic zeros of tail coefficients translate directly into missing exponential sectors of the inverse chart.

5 Complex resonance, accumulating critical values, and inverse folds

The real crossover profile

$$h(\lambda) = \frac{1-e^{-\lambda}}{\lambda}, \qquad h(0) = 1,$$

has no real zeros, but its nonzero complex zeros are $\lambda_k = 2\pi ik$, $k \in \mathbb{Z} \setminus \{0\}$. At these values an inverse expansion obtained by dividing by $h(\lambda)$ cannot remain uniform. We determine the replacement for the endpoint defect

$$F_\varepsilon(t) = \zeta(2+\varepsilon) - \mathrm{Li}_{2+\varepsilon}(e^{-t}). \tag{5.1}$$

The effect is a square-root splitting, and its geometric origin is an infinite sequence of critical values accumulating at the endpoint. The results concern compositional inversion, rather than the classical functional relation sometimes called the polylogarithm inversion formula.

5.1 An exact analytic normal form

Our convention for the Stieltjes constants is

$$\zeta(1+w) = \frac{1}{w} + \sum_{j=0}^{\infty} \frac{(-1)^j \gamma_j}{j!} w^j, \qquad \gamma_0 = \gamma. \tag{5.2}$$

Put

$$G(w) = \zeta(1+w) - \frac{1}{w}, \qquad Z(w) = w\zeta(1+w) = 1 + wG(w), \tag{5.3}$$

$$R(w) = \frac{\Gamma(1-w)}{1+w}, \tag{5.4}$$

$$A(w,t) = \sum_{j\geq 1} \frac{(-t)^j}{j!}\zeta(1+w-j), \qquad E(w,t) = \sum_{j\geq 1} \frac{(-t)^j}{(j+1)!}\zeta(1+w-j). \tag{5.5}$$

These functions have holomorphic extensions to a neighborhood of their origins. The two tails are jointly holomorphic for $|w| < r < 1$ and $|t| < 2\pi$ on smaller product domains. For example, applying the zeta functional equation to their coefficients gives a common bound $Cj^C(2\pi)^{-j}$ after division by $j!$, uniformly on a closed small w-disc. The first terms are

$$A(0,t) = \frac{t}{2} - \frac{t^2}{24} + O(t^4), \qquad E(0,t) = \frac{t}{4} - \frac{t^2}{72} + O(t^4). \tag{5.6}$$

Jonquière's convergent expansion [4], with its two poles paired before taking $\varepsilon = 0$, gives

$$F_\varepsilon(t) = t\left\{\frac{1 - R(\varepsilon)t^\varepsilon}{\varepsilon} + G(\varepsilon) + E(\varepsilon,t)\right\}, \tag{5.7}$$

$$F'_\varepsilon(t) = \mathrm{Li}_{1+\varepsilon}(e^{-t}) = \frac{1 - \Gamma(1-\varepsilon)t^\varepsilon}{\varepsilon} + G(\varepsilon) + A(\varepsilon,t). \tag{5.8}$$

Here $t^\varepsilon = \exp(\varepsilon \operatorname{Log} t)$ on the chosen logarithm branch. In the principal-sheet applications below t lies in the right half-plane near a positive real number, so the principal logarithm and the principal polylogarithm suffice. In particular, $|e^{-t}| < 1$; there is no ambiguity from analytic continuation of the defining polylogarithm series.

We shall use the constants

$$r_1 = \gamma - 1, \qquad r_2 = \frac{(\gamma-1)^2 + \zeta(2) + 1}{2}, \tag{5.9}$$

$$C = \frac{\gamma^2 + \zeta(2)}{2} + \gamma_1, \qquad D = r_2 + \gamma_1 = C - r_1. \tag{5.10}$$

Thus $R(w) = 1 + r_1 w + r_2 w^2 + O(w^3)$. Numerically, $C = 0.9162401498\ldots$ and $D = 1.3390244849\ldots$, but no numerical assumption is needed for their use.

Lemma 5.1 (Nonvanishing of the splitting constant)**.** *The constant D is strictly positive. More explicitly,*

$$D > 1 - e^{-1} > 0.$$

Proof. Euler summation gives, first for $\Re s > 1$ and then by continuation,

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} - s\int_1^\infty (\{x\} - \tfrac{1}{2}) x^{-s-1}\, dx.$$

The integral and its first s derivative converge absolutely in a neighborhood of $s = 1$. Comparing the coefficient of $s - 1$ with (5.2) proves

$$\gamma_1 = \int_1^\infty (\{x\} - \tfrac{1}{2}) \frac{1 - \log x}{x^2}\, dx.$$

Consequently

$$|\gamma_1| \le \frac{1}{2}\int_0^\infty |1 - v| e^{-v}\, dv = e^{-1}.$$

Since $\zeta(2) > 1$, formula (5.9) gives $r_2 > 1$, and the claimed lower bound follows. □

5.2 A universal chart for the critical points and critical values

Theorem 5.2 (Exponential critical-point chart)**.** *There are holomorphic functions $a(\varepsilon, \tau)$ and $b(\varepsilon, \tau)$ on one neighborhood of $(0, 0)$, independent of the integer k, such that*

$$a(\varepsilon, \tau) = 1 - C\varepsilon + \frac{\tau}{2} + O((|\varepsilon| + |\tau|)^2), \tag{5.11}$$

$$b(\varepsilon, \tau) = 1 - D\varepsilon + \frac{\tau}{4} + O((|\varepsilon| + |\tau|)^2). \tag{5.12}$$

For $k \neq 0$ and $\varepsilon \neq 0$, put

$$\tau_k = \exp(-2\pi i k/\varepsilon). \tag{5.13}$$

Whenever (ε, τ_k) lies in the chart and the logarithm is chosen so that

$$\operatorname{Log}(\tau_k a(\varepsilon, \tau_k)) = -\frac{2\pi i k}{\varepsilon} + \log a(\varepsilon, \tau_k), \tag{5.14}$$

the quantities

$$t_{c,k} = \tau_k a(\varepsilon, \tau_k), \qquad y_{c,k} = \tau_k b(\varepsilon, \tau_k) \tag{5.15}$$

are respectively a simple critical point of F_ε and its critical value. The chart contains the unique critical point for which t/τ_k is sufficiently close to 1.

At $\tau = 0$ the coefficient functions are explicit:

$$a(\varepsilon, 0) = \exp\left\{\frac{\log Z(\varepsilon) - \log\Gamma(1 - \varepsilon)}{\varepsilon}\right\}, \tag{5.16}$$

$$b(\varepsilon, 0) = a(\varepsilon, 0)\frac{Z(\varepsilon)}{1 + \varepsilon}. \tag{5.17}$$

The quotient in (5.16) is holomorphic at zero and vanishes there. Thus these formulas determine every algebraic-order coefficient in ε of the leading exponential sector. The first coefficients of the next exponential sector are also explicit:

$$\partial_\tau a(\varepsilon, 0) = -\frac{\zeta(\varepsilon) a(\varepsilon, 0)^2}{Z(\varepsilon)}, \qquad \partial_\tau b(\varepsilon, 0) = -\frac{\zeta(\varepsilon) a(\varepsilon, 0)^2}{2}. \tag{5.18}$$

Proof. For u near 1, use the analytic logarithm with $\log 1 = 0$ and define

$$\mathcal{H}(\varepsilon, \tau, u) = \frac{1 - \Gamma(1 - \varepsilon) e^{\varepsilon \log u}}{\varepsilon} + G(\varepsilon) + A(\varepsilon, \tau u). \tag{5.19}$$

The numerator in the quotient vanishes at $\varepsilon = 0$, so $\mathcal{H}$ is jointly holomorphic. Its specialization at $(\varepsilon, \tau) = (0, 0)$ is $-\log u$, whose derivative at $u = 1$ is -1. The holomorphic implicit function theorem therefore gives the unique $u = a(\varepsilon, \tau)$ near 1. The derivative with respect to u remains nonzero on a smaller neighborhood.

Under (5.14), the identity $t_{c,k}^{\varepsilon} = a(\varepsilon, \tau_k)^{\varepsilon}$ holds exactly, since $e^{-2\pi ik} = 1$. Equation (5.8) now identifies $\mathcal{H} = 0$ with $F_{\varepsilon}'(t_{c,k}) = 0$. Moreover,

$$F_{\varepsilon}''(t_{c,k}) = \tau_k^{-1} \partial_u \mathcal{H}(\varepsilon, \tau_k, a(\varepsilon, \tau_k)) \neq 0,$$

which proves simplicity. Expanding (5.19) to first order at $(0, 0, 1)$ gives

$$\mathcal{H} = -(u - 1) - C\varepsilon + \frac{\tau}{2} + O((|u - 1| + |\varepsilon| + |\tau|)^2).$$

This proves (5.11).

At a critical point, (5.8) implies

$$\frac{\Gamma(1 - \varepsilon)t^{\varepsilon}}{\varepsilon} = \zeta(1 + \varepsilon) + A(\varepsilon, t).$$

Substitution in (5.7) gives the regular expression

$$\frac{F_{\varepsilon}(t)}{t} = \frac{Z(\varepsilon)}{1 + \varepsilon} + E(\varepsilon, t) - \frac{A(\varepsilon, t)}{1 + \varepsilon}.$$

Define b by multiplying this right-hand side, evaluated at $t = \tau a(\varepsilon, \tau)$, by $a(\varepsilon, \tau)$. It is holomorphic. Equations (5.6) and (5.11) show that its linear terms are $-D\varepsilon + \tau/4$. When $\tau = 0$, the critical equation is $\Gamma(1 - \varepsilon)a^{\varepsilon} = Z(\varepsilon)$. The branches near $a = 1$ give (5.16), and then (5.17). The linear terms of the two logarithms in (5.16) are both $\gamma\varepsilon$, proving the asserted removability and value. At $\tau = 0$ the derivatives of the critical equation are $\mathcal{H}_u = -Z(\varepsilon)/a(\varepsilon, 0)$ and $\mathcal{H}_{\tau} = -\zeta(\varepsilon)a(\varepsilon, 0)$. Implicit differentiation gives the first identity in (5.18). Differentiating the displayed regular critical-value expression, and using $A_t(\varepsilon, 0) = -\zeta(\varepsilon)$ and $E_t(\varepsilon, 0) = -\zeta(\varepsilon)/2$, gives the second. □

Corollary 5.3 (Infinitely many principal-sheet critical values)**.** *Fix a sufficiently small nonzero real β and take $\varepsilon = i\beta$. For integers k with $k\beta > 0$, and $|k|$ sufficiently large, all the points in (5.15) belong to the principal polylogarithm sheet. They are distinct simple critical points, and their nonzero critical values are distinct for sufficiently large $|k|$. If they are indexed in the direction $|k| \to \infty$, then*

$$\frac{t_{c,k+\operatorname{sgn}\beta}}{t_{c,k}} \longrightarrow e^{-2\pi/|\beta|}, \qquad \frac{y_{c,k+\operatorname{sgn}\beta}}{y_{c,k}} \longrightarrow e^{-2\pi/|\beta|}. \tag{5.20}$$

Each $y_{c,k}$ is a square-root branch value of a local compositional inverse of $F_{i\beta}$. Equivalently, the inverse of $z \mapsto \mathrm{Li}_{2+i\beta}(z)$ has square-root branch values $\zeta(2 + i\beta) - y_{c,k}$ accumulating at $\zeta(2 + i\beta)$.

Proof. Here $\tau_k = e^{-2\pi k/\beta}$ is positive. Since a is uniformly close to 1, $t_{c,k}$ lies in the right half-plane and (5.14) holds for the principal logarithm. Equations (5.16)–(5.17) give nonzero limits $a(i\beta, 0)$ and $b(i\beta, 0)$. The geometric ratios and eventual distinctness follow. Taylor expansion at a simple critical point gives

$$F_{i\beta}(t) = y_{c,k} + \tfrac{1}{2}F_{i\beta}''(t_{c,k})(t - t_{c,k})^2 + O((t - t_{c,k})^3).$$

An analytic change of the local t coordinate reduces this exactly to a nonzero multiple of a square. Its two inverses are exchanged by a loop around $y_{c,k}$. Finally $z = e^{-t}$ is locally biholomorphic, so the same ramification holds for the polylogarithm inverse. □

Proposition 5.4 (Exhaustion of the small principal-sheet critical points)**.** *For every sufficiently small fixed nonzero real β, all sufficiently small critical points of $F_{i\beta}$ in the principal slit $0 < |t| < r$, $-\pi < \arg t < \pi$, belong to the sequence $t_{c,k}$ of Theorem 5.2. Each such point has exactly one index k, and that index satisfies $k\beta > 0$. Consequently the chart classifies all sufficiently small principal-sheet zeros of $\mathrm{Li}_{1+i\beta}(e^{-t})$, not merely a subsequence.*

Proof. On a sufficiently small product neighborhood define

$$Q(\varepsilon, t) = \frac{\log Z(\varepsilon) - \log \Gamma(1-\varepsilon)}{\varepsilon} + \frac{1}{\varepsilon} \log\left(1 + \frac{\varepsilon A(\varepsilon, t)}{Z(\varepsilon)}\right), \tag{5.21}$$

where every logarithm is the holomorphic branch near 1. The first numerator has a double zero at $\varepsilon = 0$; the second quotient is also removable, by the analytic germ $\log(1+w)/w$. Thus Q is jointly holomorphic and $Q(0,0) = 0$.

If $F'_\varepsilon(t) = 0$ and $\varepsilon \neq 0$, then (5.8) gives

$$\Gamma(1-\varepsilon) e^{\varepsilon \operatorname{Log} t} = Z(\varepsilon) + \varepsilon A(\varepsilon, t).$$

The right side is nonzero on the chosen neighborhood. Taking logarithms of this equality of exponentials gives a unique integer n such that

$$\operatorname{Log} t = Q(\varepsilon, t) + \frac{2\pi i n}{\varepsilon}.$$

For $\varepsilon = i\beta$ and $k = -n$, this becomes

$$\operatorname{Log} t = Q(i\beta, t) - \frac{2\pi k}{\beta}, \qquad t = \tau_k \exp Q(i\beta, t), \quad \tau_k = e^{-2\pi k/\beta} > 0. \tag{5.22}$$

Shrink the neighborhoods so that $\exp Q$ lies inside the uniqueness neighborhood of $u = 1$ in Theorem 5.2. Because Q is bounded, $|t|$ sufficiently small forces $k/\beta > 0$ and ensures that τ_k is in the chart. Moreover $\arg t = \operatorname{Im} Q(i\beta, t)$ is small, so the principal logarithm satisfies the compatibility condition of that theorem. Its uniqueness statement now gives $t = \tau_k a(i\beta, \tau_k) = t_{c,k}$. The integer in (5.22) is unique, which proves the claimed uniqueness of the index. For fixed β, a sequence with $t \to 0$ necessarily has $|k| \to \infty$. □

The geometric accumulation mechanism follows from the familiar balance between a constant term and t^ε in Jonquière's expansion. The useful refinement here is one jointly holomorphic chart at $(\varepsilon, \tau) = (0,0)$, together with the exact critical-value chart and the inverse transition proved next. The exhaustion statement applies to all sufficiently small zeros in the stated small imaginary-order regime. It does not classify zeros away from the endpoint or assert a corresponding accumulation for arbitrary nonreal ε on a fixed principal sheet.

Proposition 5.5 (The necessary line for principal-sheet accumulation)**.** *Fix $s \in \mathbb{C}$. If $\Re s \neq 1$, or if $s = 1$, then the principal branch of $\mathrm{Li}_s(e^{-t})$ has no zeros in a sufficiently small punctured slit neighborhood $0 < |t| < r$, $-\pi < \arg t < \pi$. Consequently a principal-sheet sequence of zeros accumulating at $t = 0$ is possible only for $\Re s = 1$ with $s \neq 1$. The small imaginary-displacement family in Corollary 5.3 lies on exactly this necessary line.*

Proof. If $\Re s < 1$, Jonquière's formula gives

$$\mathrm{Li}_s(e^{-t}) = \Gamma(1-s) t^{s-1} + O(1)$$

uniformly in the principal slit. Here $\Gamma(1-s) \neq 0$, and

$$|t^{s-1}| = |t|^{\Re s - 1} e^{-\Im s \arg t} \geq e^{-\pi |\Im s|} |t|^{\Re s - 1} \longrightarrow \infty.$$

The leading term therefore cannot be canceled. If $\Re s > 1$, the same formula yields $\mathrm{Li}_s(e^{-t}) \to \zeta(s)$ uniformly in the slit. At positive integer $s \geq 2$ the paired term is a multiple of $t^{s-1}\operatorname{Log} t$ and also tends to zero. The absolutely convergent Euler product implies $\zeta(s) \neq 0$ for $\Re s > 1$, so there are again no zeros sufficiently near zero. Finally $\mathrm{Li}_1(e^{-t}) = -\operatorname{Log}(1-e^{-t}) = -\operatorname{Log} t + O(t)$, whose modulus tends to infinity uniformly in the principal slit. □

This restriction gives a precise correction to an external heuristic claim. Vepstas's 2023 draft [18] describes an infinite sequence of zeros of $\mathrm{Li}_{1/2+i\tau}$ accumulating at $z = 1$ on the displayed principal sheet. Proposition 5.5 rules out that asymptotic claim for each fixed τ. The modulus calculation used there is not valid: for $q = re^{i\theta}$ and $s = \sigma + i\tau$,

$$|q^{-s}| = r^{-\sigma}e^{\tau\theta}, \qquad \text{not } r^{\tau\theta-\sigma}.$$

For fixed τ, the factor $e^{\tau\theta}$ remains bounded on the principal slit, while $r^{-\sigma}$ has its stated unbounded or vanishing power behavior. This is a correction to that external draft, not a correction attributed to the inspected repository. It does not exclude many finite zeros near the branch point, a joint large-$|\tau|$ limit, or zero sequences on varying logarithm sheets.

5.3 The two inverse branches at a zero of the crossover profile

Fix $k \neq 0$, abbreviate $\lambda_0 = 2\pi i k$, and write

$$Y = \log(1/y), \qquad x = Y^{-1}, \qquad \rho = Y^{-1/2}, \qquad \chi = yY, \qquad y > 0. \tag{5.23}$$

Theorem 5.6 (Convergent square-root–exponential inverse chart)**.** *For all sufficiently large real Y, put $y = e^{-Y}$ and $\varepsilon = \lambda_0/Y$. The equation*

$$\zeta(2+\varepsilon) - \mathrm{Li}_{2+\varepsilon}(e^{-t}) = y \tag{5.24}$$

has exactly two roots, counted with multiplicity, in a fixed sufficiently small disc $|t/y - 1| < \delta$. They are simple. There exist holomorphic functions V_+ and V_- on a neighborhood of $(0,0)$ such that these roots are exactly

$$t_\pm = y\left[1 + \rho V_\pm(\rho, \chi)\right], \qquad V_\pm(0,0) = c_\pm := \pm\sqrt{-2\lambda_0 D}. \tag{5.25}$$

Either initial square root may be labeled $+$; the other is its negative. In particular,

$$\frac{t_\pm}{y} = 1 + \frac{c_\pm}{\sqrt{Y}} - \frac{\lambda_0(r_1 + 4D/3)}{Y} + O(Y^{-3/2}). \tag{5.26}$$

The full series in the two independent variables ρ and χ is convergent. Its first coefficients satisfy

$$\partial_\rho V_\pm(0,0) = -\lambda_0(r_1 + 4D/3), \qquad \partial_\chi V_\pm(0,0) = \frac{1}{4c_\pm}. \tag{5.27}$$

Thus the analytic tail contributes a correction of leading size $y\sqrt{Y}/(4c_\pm)$ to $t_\pm/y$, although the algebraic corrections are in descending half-powers of Y.

Proof. Put $t = yu$. Since $\varepsilon Y = \lambda_0$ and $e^{-\lambda_0} = 1$, equation (5.7), divided by y, becomes

$$\Phi_0(x, y, u) = u\left\{\frac{1 - R(\lambda_0 x)e^{\lambda_0 x \log u}}{\lambda_0 x} + G(\lambda_0 x) + E(\lambda_0 x, yu)\right\} - 1 = 0. \tag{5.28}$$

Regard x, y, u as independent variables. The quotient is holomorphic at $x = 0$, and

$$\Phi_0(0, 0, u) = u(1 - \log u) - 1 = -\tfrac{1}{2}(u-1)^2 + \tfrac{1}{6}(u-1)^3 + O((u-1)^4). \tag{5.29}$$

The root at $u = 1$ is exactly double. The coefficient of x at $u = 1$ is $-\lambda_0 D$, while the coefficient of y is $1/4$.

After $x = \rho^2$, $y = \rho^2\chi$, and $u = 1 + \rho v$, the function

$$\mathcal{K}(\rho, \chi, v) = \rho^{-2}\Phi_0(\rho^2, \rho^2\chi, 1 + \rho v) \tag{5.30}$$

is holomorphic at $\rho = 0$. Indeed the constant and linear powers of ρ vanish identically in the convergent Taylor expansion. Its limiting equation is

$$\mathcal{K}(0, \chi, v) = -\frac{v^2}{2} - \lambda_0 D + \frac{\chi}{4}. \tag{5.31}$$

At $\chi = 0$ it has the two distinct roots $c_\pm$, by Lemma 5.1. The v derivative is $-c_\pm \neq 0$. The holomorphic implicit function theorem gives the two functions $V_\pm$. Their Taylor series converge in one polydisc; the physical arguments $\rho = Y^{-1/2}$ and $\chi = Ye^{-Y}$ lie in that polydisc for sufficiently large Y.

Choose $\delta > 0$ so small that (5.29) has no zero in $0 < |u - 1| \leq \delta$. Uniform convergence on the boundary as $(x, y) \to (0, 0)$ and Rouché's theorem imply that the perturbed equation has exactly two roots in the disc, counted with multiplicity. The constructed roots have different leading coefficients and are simple because $\partial_v \mathcal{K}$ stays nonzero. This proves the exact root count and (5.25).

For the coefficients, writing $w = u - 1$ gives

$$\begin{aligned}\Phi_0(x, 0, 1 + w) = &-\frac{w^2}{2} + \frac{w^3}{6} + O(w^4)\\ &- \lambda_0 x\left[D + (D + r_1)w + O(w^2)\right] + O(x^2).\end{aligned}$$

Insert $w = c\rho + d\rho^2 + O(\rho^3)$. The ρ^2 coefficient is $-c^2/2 - \lambda_0 D$. The ρ^3 coefficient is $-cd + c^3/6 - \lambda_0(D + r_1)c$. Solving gives $c = c_\pm$ and $d = -\lambda_0(r_1 + 4D/3)$. Differentiating (5.31) at $\chi = 0$ gives $\partial_\chi V = 1/(4c)$. Finally $\chi = Ye^{-Y}$ is smaller than every inverse power of Y, which justifies the algebraic remainder in (5.26). □

Remark 5.7 (Transseries support and local scope). Writing $V_\pm(\rho, \chi) = \sum_{a,b\geq 0} v^\pm_{a,b}\rho^a\chi^b$ in (5.25) gives the convergent expansion

$$t_\pm = y + \sum_{a,b\geq 0} v^\pm_{a,b} y^{b+1} Y^{\,b-(a+1)/2}.$$

The half-power scale is forced by a genuine double root, rather than by a choice of notation for an ordinary integer-power series. The theorem counts roots only in the stated disc near $t = y$. It makes no global uniqueness assertion for the polylogarithm inverse.

Corollary 5.8 (One inverse chart for all exact lattice resonances)**.** *There are two holomorphic functions $W_\pm(\sigma, z)$ on one neighborhood of $(0, 0)$, independent of any resonance index, such that*

$$W_\pm(0, 0) = \pm\sqrt{-2D}, \tag{5.32}$$

$$\partial_\sigma W_\pm(0, 0) = -(r_1 + 4D/3), \qquad \partial_z W_\pm(0, 0) = \frac{1}{4W_\pm(0, 0)}. \tag{5.33}$$

Let $y = e^{-Y} > 0$, $Y > 0$, and suppose $\varepsilon Y \in 2\pi i\mathbb{Z} \setminus \{0\}$. Choose $\sigma^2 = \varepsilon$ and set $z = y/\varepsilon$. Whenever $|\sigma|$ and $|z|$ are sufficiently small, the two simple roots near $t = y$ are exactly

$$t_\pm = y\{1 + \sigma W_\pm(\sigma, y/\varepsilon)\}. \tag{5.34}$$

The common neighborhood and its local root-count statement do not depend on the integer in the resonance condition. The square-root choice fixes the labeling; the unordered pair of roots is unchanged by a coherent change of that choice.

Proof. For independent variables near $(\varepsilon, y, u) = (0, 0, 1)$ put

$$\Psi(\varepsilon, y, u) = u\left\{\frac{1 - R(\varepsilon)e^{\varepsilon \log u}}{\varepsilon} + G(\varepsilon) + E(\varepsilon, yu)\right\} - 1.$$

The quotient is removable, so Ψ is holomorphic. Under the physical resonance condition and $t = yu$ near $y > 0$, the principal logarithm gives $t^\varepsilon = e^{-\varepsilon Y}u^\varepsilon = u^\varepsilon$. Thus $\Psi = 0$ is the original inverse equation divided by y.

Its leading terms at $u = 1 + w$ are

$$\Psi(\varepsilon, y, 1 + w) = -\frac{w^2}{2} + \frac{w^3}{6} - \varepsilon\{D + (D + r_1)w\} + \frac{y}{4} + \text{higher terms}.$$

The convergent substitution

$$\widetilde{K}(\sigma, z, w) = \sigma^{-2}\Psi(\sigma^2, \sigma^2 z, 1 + \sigma w)$$

is holomorphic at $\sigma = 0$, since the constant and linear powers of σ vanish. Its limit is

$$\widetilde{K}(0, z, w) = -\frac{w^2}{2} - D + \frac{z}{4}.$$

At $z = 0$ its two roots are the nonzero numbers in (5.32). The holomorphic implicit function theorem produces $W_\pm$ on one common neighborhood. The same local Rouché argument as in Theorem 5.6 proves that they supply precisely the two roots in a fixed sufficiently small disc about $u = 1$; they are simple for $\sigma \neq 0$. Comparing the cubic terms gives $\partial_\sigma W = -(r_1 + 4D/3)$, and differentiating the limiting equation in z gives the other identity in (5.33). Every construction in this proof is independent of the resonance index. □

Recovery of the fixed-index coordinates. With $\lambda_0 = 2\pi ik$, $\rho = Y^{-1/2}$, and $\chi = Ye^{-Y}$, choose compatible square roots. Then

$$V_\pm(\rho, \chi) = \sqrt{\lambda_0}\, W_\pm\left(\sqrt{\lambda_0}\rho, \frac{\chi}{\lambda_0}\right).$$

In particular, the universal chart permits a growing resonance index provided the actual coordinates $\sigma = \sqrt{\varepsilon}$ and y/ε remain in its stated small domain. It makes no uniform claim outside those coordinate conditions.

5.4 The parameter unfolding and its exact branch locus

The small order displacement that changes the leading square-root coefficient is smaller than the original Y^{-1} spectral scale. Specifically, write

$$\lambda = \lambda_0 + \frac{\eta}{Y^2}, \qquad \varepsilon = \frac{\lambda}{Y} = \frac{\lambda_0}{Y} + \frac{\eta}{Y^3}. \tag{5.35}$$

Theorem 5.9 (Uniform unfolding away from its limiting fold)**.** *Let $\eta_* = \lambda_0^2 D$. On any simply connected bounded domain of η whose closure avoids η_*, choose either branch of*

$$c_\pm(\eta) = \pm\sqrt{\frac{2\eta}{\lambda_0} - 2\lambda_0 D}.$$

Locally uniformly in η on that domain, the two roots near $t = y$ of (5.24), with (5.35), have exact representations

$$t_\pm = y\left[1 + Y^{-1/2}V_\pm\left(\eta, Y^{-1/2}, Ye^{-Y}\right)\right], \qquad V_\pm(\eta, 0, 0) = c_\pm(\eta), \tag{5.36}$$

where $V_\pm$ are jointly holomorphic in their arguments on suitable neighborhoods of compact subsets. Their first correction is

$$\frac{t_\pm}{y} = 1 + \frac{c_\pm(\eta)}{\sqrt{Y}} + \frac{4\eta/(3\lambda_0) - \lambda_0(r_1 + 4D/3)}{Y} + O(Y^{-3/2}). \tag{5.37}$$

The exact local branch locus is described by the next theorem; its leading scaled location is η_*.

Proof. Introduce independent variables x, α, y and put $\lambda = \lambda_0 + x\alpha$. Define

$$\Phi(x, \alpha, y, u) = u\left\{\frac{1 - R(\lambda x)e^{-x\alpha}e^{\lambda x \log u}}{\lambda x} + G(\lambda x) + E(\lambda x, yu)\right\} - 1. \tag{5.38}$$

The numerator of the quotient vanishes identically at $x = 0$; since $\lambda_0 \neq 0$, division by λx is removable. Thus Φ is holomorphic near $(x, \alpha, y, u) = (0, 0, 0, 1)$, with

$$\Phi(0, \alpha, 0, u) = u\left(1 - \log u + \frac{\alpha}{\lambda_0}\right) - 1. \tag{5.39}$$

On the physical locus $x = 1/Y$, $y = e^{-Y}$, this is precisely the original inverse equation after setting $t = yu$. For (5.35), take $\alpha = \eta\rho^2$ and $x = \rho^2$. Then

$$\rho^{-2}\Phi(\rho^2, \eta\rho^2, \rho^2\chi, 1 + \rho v) \longrightarrow -\frac{v^2}{2} - \lambda_0 D + \frac{\eta}{\lambda_0} + \frac{\chi}{4}.$$

The implicit function theorem gives (5.36), and its uniformity follows on compact subsets where $c_\pm(\eta) \neq 0$. For the next coefficient, the additional term $(\eta/\lambda_0)xu$ contributes $(\eta/\lambda_0)c\rho^3$. Therefore

$$d = \frac{c^2}{6} - \lambda_0(D + r_1) + \frac{\eta}{\lambda_0} = \frac{4\eta}{3\lambda_0} - \lambda_0(r_1 + 4D/3).$$

The same Rouché argument as in Theorem 5.6 proves that these are the two roots in the chosen local disc. □

Theorem 5.10 (Exact analytic fold and local monodromy)**.** *For the function (5.38) there are unique holomorphic functions $u_c(x, y)$ and $\alpha_c(x, y)$ near $(0, 0)$ such that*

$$\Phi(x, \alpha_c(x, y), y, u_c(x, y)) = 0, \qquad \partial_u\Phi(x, \alpha_c(x, y), y, u_c(x, y)) = 0.$$

Their first terms are

$$\alpha_c(x, y) = \lambda_0^2 D\, x - \frac{\lambda_0}{4}y + O((|x| + |y|)^2), \tag{5.40}$$

$$u_c(x, y) = 1 - \lambda_0 r_1 x + \frac{y}{4} + O((|x| + |y|)^2). \tag{5.41}$$

Consequently the exact scaled-order branch locus on the physical curve is

$$\lambda_c(Y) = \lambda_0 + \frac{1}{Y}\alpha_c\left(\frac{1}{Y}, e^{-Y}\right) = \lambda_0 + \frac{\lambda_0^2 D}{Y^2} - \frac{\lambda_0 e^{-Y}}{4Y} + O\left(\frac{1}{Y}\left(\frac{1}{Y} + e^{-Y}\right)^2\right). \tag{5.42}$$

Near this locus, a holomorphic change of the u coordinate gives the exact normal form

$$\Phi = (\alpha - \alpha_c(x, y))\mathcal{A}(x, \alpha, y) - \frac{v^2}{2}, \qquad \mathcal{A}(0, 0, 0) = \frac{1}{\lambda_0}. \tag{5.43}$$

Thus there are two local inverse branches off the locus, a double root on it, and analytic continuation once around the locus exchanges the two branches.

Proof. At the base point the relevant derivatives are

$$\Phi_u = 0, \quad \Phi_{uu} = -1, \quad \Phi_\alpha = \lambda_0^{-1}, \quad \Phi_x = -\lambda_0 D, \quad \Phi_y = \tfrac{1}{4}.$$

The Jacobian of (Φ, Φ_u) with respect to (α, u) is invertible. The holomorphic implicit function theorem gives the two critical functions. In addition,

$$\Phi_{ux} = -\lambda_0(D + r_1), \qquad \Phi_{uy} = \tfrac{1}{2},$$

while $\Phi_{u\alpha} = \lambda_0^{-1}$. Differentiating the two critical equations at the origin yields (5.40)–(5.41).

For completeness, the normal form does not require an unproved appeal to a global classification theorem. Solve $\Phi_u = 0$ for $u = u_*(x, \alpha, y)$ near 1. Set $M(x, \alpha, y) = \Phi(x, \alpha, y, u_*)$. Taylor expansion in $w = u - u_*$ gives a holomorphic factor K such that

$$\Phi = M - \frac{w^2}{2} K(x, \alpha, y, w), \qquad K(0, 0, 0, 0) = 1.$$

Choose the holomorphic square root of K near 1 and define $v = w\sqrt{K}$. This is a locally invertible analytic coordinate change. Furthermore $M = 0$ exactly when $\alpha = \alpha_c(x, y)$, and $M_\alpha(0, 0, 0) = 1/\lambda_0$. Division by this simple factor gives $M = (\alpha - \alpha_c)\mathcal{A}$, proving (5.43). Its two solutions in the v coordinate are $\pm\sqrt{2(\alpha - \alpha_c)\mathcal{A}}$, with the asserted ramification and monodromy. Substitution of the physical variables gives (5.42). □

5.5 Status, comparison, and possible extensions

The pole-paired local polylogarithm expansion is classical; near-integer order expansions and order derivatives were already treated in the computational literature, including Bailey–Borwein and Roughan [9, 8]. The repository's all-integer-order resonance report [1] proves a direct uniform expansion and explicitly notes the nonzero imaginary zeros of the leading profile. Its proposed questions distinguish absolute from relative errors near those zeros. The repository's confluent critical transseries article [3] proves an exact inverse chart for a real moving exponent and a specified positive feedback model. Neither inspected statement establishes the complex square-root splitting (5.25) or the exact spectral fold (5.42).

Exponential accumulation of polylogarithm zeros has also appeared in heuristic discussions, for example Vepstas's 2023 note [18]. The present claims of contribution should therefore concern the proved common analytic chart, its critical values, its precise principal-sheet regime, and the uniform inverse unfolding. No comprehensive publication-priority claim is made. In particular, a broad claim about zero accumulation at arbitrary fixed nonreal order would require additional branch analysis and is not a consequence of the theorems above.

6 Exact Borel sums and integer-order jets at large negative argument

Write

$$F_s(x) = -\operatorname{Li}_s(-e^x), \qquad x \in \mathbb{R}, \tag{6.1}$$

with the principal polylogarithm. Our order is shifted by one from the customary Fermi–Dirac order: the function denoted $F_q(x)$ in DLMF §25.12(iii) is our $F_{q+1}(x)$. The Sommerfeld expansion and its exponentially small completion are classical; see Dingle, Garoni–Frankel–Glasser, and Gil–Segura–Temme [15, 16, 17]. The purpose here is to give an elementary exact ordinary-Borel representation, then use order differentiation at a terminating value to recover the complete exponential tower. In particular, no novelty is claimed for the Sommerfeld coefficients or for the existence of the exponentially small Fermi–Dirac correction.

Put

$$c_0 = 1, \qquad c_{2k} = 2\eta_{\mathrm{D}}(2k) = 2(1 - 2^{1-2k})\zeta(2k) \quad (k \geq 1), \tag{6.2}$$

where η_{D} is the Dirichlet eta function, and define the formal series

$$A_s(x) = 1 + \sum_{k\geq 1} a_{2k}(s)x^{-2k}, \qquad a_{2k}(s) = c_{2k}\frac{\Gamma(s+1)}{\Gamma(s+1-2k)}. \tag{6.3}$$

The quotient in this formula is a falling factorial and hence a polynomial in s. The normalized Sommerfeld series is $x^s A_s(x)/\Gamma(s+1)$. For convenience $c_j = 0$ when j is odd. We write $\psi = \Gamma'/\Gamma$ and $\psi_1 = \psi'$.

6.1 The ordinary Borel transform and its exact lateral sums

Theorem 6.1 (An explicit Borel transform and exact completion)**.** *Let*

$$\Omega = \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty)).$$

For $p \in \Omega$ use the principal branches of $1 \pm p/m$ and set

$$B_s(p) = 1 + \sum_{m=1}^{\infty} (-1)^{m-1} \left\{(1 + p/m)^s + (1 - p/m)^s - 2\right\}. \tag{6.4}$$

This series converges locally normally in $(s, p) \in \mathbb{C} \times \Omega$. It is the ordinary Borel transform of (6.3):

$$B_s(p) = 1 + \sum_{k\geq 1} \frac{a_{2k}(s)}{(2k)!}p^{2k}, \qquad |p| < 1. \tag{6.5}$$

For every real $x > 0$ and $\operatorname{Re} s > -1$, the lateral integrals

$$G_s^{\pm}(x) = \frac{x^{s+1}}{\Gamma(s+1)} \int_0^{\infty} e^{-xp} B_s(p \pm i0)\, dp \tag{6.6}$$

exist as absolutely convergent improper integrals, are holomorphic in s, and satisfy the exact identities

$$\boxed{G_s^{\pm}(x) = F_s(x) + e^{\mp i\pi s} F_s(-x).} \tag{6.7}$$

Here the upper boundary value in the p plane gives $\arg(1-p/m) = -\pi$ when $p > m$. Consequently, with the convention $\operatorname{Disc}_0 G_s = G_s^+ - G_s^-$,

$$\operatorname{Disc}_0 G_s(x) = -2i \sin(\pi s) F_s(-x). \tag{6.8}$$

The arithmetic average $M_s = (G_s^+ + G_s^-)/2$ obeys

$$F_s(x) = M_s(x) - \cos(\pi s) F_s(-x). \tag{6.9}$$

All these identities may be differentiated any finite number of times with respect to s locally uniformly in $\operatorname{Re} s > -1$.

Proof. On compact subsets of $\mathbb{C} \times \Omega$, the expression in braces in (6.4) is $O(m^{-2})$. The linear Taylor terms cancel. This proves local normal convergence, including all order derivatives. For $|p| < 1$ the binomial expansion and absolute convergence give

$$\sum_{m\geq 1} (-1)^{m-1} \left\{(1 + p/m)^s + (1 - p/m)^s - 2\right\} = 2 \sum_{k\geq 1} \binom{s}{2k} \eta_{\mathrm{D}}(2k) p^{2k},$$

which is (6.5).

We justify the integral interchange explicitly, since the paired expression in (6.4) is essential. Fix $x > 0$ and write

$$h_s^{\pm}(y) = (1 + y/x)^s + (1 - y/x \mp i0)^s - 2.$$

After the substitution $y = xp/m$, the sum of the integrals of the absolute values of the paired terms is

$$\sum_{m\geq 1} \int_0^\infty xe^{-xp} \left|(1 + p/m)^s + (1 - p/m \mp i0)^s - 2\right| dp = \int_0^\infty \frac{e^{-y}}{(1 - e^{-y})^2} |h_s^{\pm}(y)|\, dy. \tag{6.10}$$

Near $y = 0$, $h_s^{\pm}(y) = O(y^2)$; near $y = x$ its only possible unbounded contribution is a constant times $|y - x|^{\operatorname{Re} s}$; and at infinity it has at most polynomial growth. Thus the last integral is finite exactly in the domain needed here, $\operatorname{Re} s > -1$. The same majorant works locally uniformly in s, with powers of $|\log|y - x||$ for order derivatives. This proves Fubini interchange, absolute convergence, and holomorphy of the lateral integrals.

Let

$$\rho(y) = \frac{e^{-y}}{(1 + e^{-y})^2} = \frac{1}{4\cosh^2(y/2)}.$$

Summing the alternating exponential series gives $\sum_{m\geq 1}(-1)^{m-1} me^{-my} = \rho(y)$ for $y > 0$. Since $\int_0^\infty \rho(y)dy = 1/2$, (6.6) becomes

$$G_s^{\pm}(x) = \frac{1}{\Gamma(s + 1)} \int_0^\infty \rho(y) \left\{(x + y)^s + (x - y \mp i0)^s\right\} dy. \tag{6.11}$$

For $\operatorname{Re} s > 0$, integration by parts in the standard Fermi–Dirac integral gives

$$F_s(v) = \frac{1}{\Gamma(s + 1)} \int_0^\infty t^s \rho(t - v)dt, \qquad v \in \mathbb{R}. \tag{6.12}$$

Both sides are holomorphic for $\operatorname{Re} s > -1$ after continuation in the order, so the formula holds throughout that half-plane. The portion of (6.11) with $y < x$, together with the $(x + y)^s$ integral, is $F_s(x)$. In the remaining portion put $t = y - x$. Its value is $e^{\mp i\pi s} F_s(-x)$. This proves (6.7); subtraction and averaging prove the other formulas. □

Remark 6.2 (Asymptotics and uniform order differentiation). Taylor expansion of B_s on $0 \leq p \leq \delta < 1$ and elementary Laplace integration prove the Sommerfeld expansion with a Poincaré remainder, locally uniformly in $\operatorname{Re} s > -1$. The integral over $p \geq \delta$ is exponentially small: for $x \geq x_0 > 0$ its absolute value is bounded by $e^{-(x-x_0)\delta} \int_\delta^\infty e^{-x_0 p} |B_s(p \pm i0)| dp$. The same argument applies to all finite order jets. In particular, the derivative of the asymptotic expansion is justified by uniform holomorphy, not by an interchange with an unexamined divergent series. The arithmetic average in (6.9) is a linear symmetric sum; no nonlinear median-summation convention is being imposed.

Proposition 6.3 (Local singularities and termination)**.** *On a sheet obtained by continuation along a path avoiding the nonzero integers, the local expression at $p = m \in \mathbb{N}$ is*

$$B_s(p) = H_{m,s}(p) + (-1)^{m-1}(1 - p/m)^s, \tag{6.13}$$

where $H_{m,s}$ is holomorphic in p near m and holomorphic in s. There is an analogous expression at $p = -m$, with $1 + p/m$. For $N \in \mathbb{Z}_{\geq 0}$, B_N is a polynomial of degree $2\lfloor N/2\rfloor$. For every $r \geq 1$, the singular part of its rth order jet at $s = N$ is

$$(-1)^{m-1}(1 - p/m)^N \log^r(1 - p/m). \tag{6.14}$$

Proof. Only the mth summand in (6.4) can have a singularity at m; the other summands continue holomorphically to a sufficiently small neighborhood on the chosen sheet. Local normal convergence handles the infinite tail. The binomial theorem proves polynomial termination, and order differentiation gives (6.14). □

6.2 Recovering exponential sectors from order jets

At an integer $N \geq 0$, define the finite polynomial

$$P_N(x) = \sum_{0 \leq 2k \leq N} c_{2k} \frac{x^{N-2k}}{(N-2k)!}. \tag{6.15}$$

The exact specialization of Theorem 6.1 is the classical integer reflection identity

$$F_N(x) = P_N(x) - (-1)^N F_N(-x), \qquad F_N(-x) = \sum_{m \geq 1} \frac{(-1)^{m-1} e^{-mx}}{m^N}. \tag{6.16}$$

The formal algebraic series terminates, while the second series in (6.16) is nonzero. The entire exponential tower is invisible to the single specialized polynomial B_N.

Theorem 6.4 (Triangular reconstruction from the order jets)**.** *For $N, r \in \mathbb{Z}_{\geq 0}$, put*

$$E_{N,r}(x) = \partial_s^r F_s(-x)|_{s=N} = \sum_{m \geq 1} \frac{(-1)^{m-1} (-\log m)^r}{m^N} e^{-mx}, \tag{6.17}$$

$$D_{N,r}(x) = \partial_s^r (G_s^+(x) - G_s^-(x))\Big|_{s=N}. \tag{6.18}$$

For $r = 0$ the convention $(-\log 1)^0 = 1$ is used. Then $D_{N,0} = 0$, and, for $r \geq 1$,

$$D_{N,r} = -2i(-1)^N \sum_{j=0}^{\lfloor (r-1)/2 \rfloor} \binom{r}{2j+1} (-1)^j \pi^{2j+1} E_{N,r-2j-1}. \tag{6.19}$$

In particular,

$$\boxed{F_N(-x) = \frac{(-1)^{N+1}}{2\pi i} D_{N,1}(x).} \tag{6.20}$$

Every $E_{N,r}$ is uniquely recovered from the discontinuities through order $r+1$. Equivalently, as an identity of analytic germs at $\varepsilon = 0$,

$$\sum_{r \geq 0} \frac{E_{N,r}}{r!} \varepsilon^r = \frac{(-1)^{N+1}}{2i \sin(\pi\varepsilon)} \sum_{r \geq 1} \frac{D_{N,r}}{r!} \varepsilon^r. \tag{6.21}$$

The apparent singularity on the right is removable.

Proof. The exponentially convergent series for $F_s(-x)$ is locally uniformly convergent for every complex s, and hence may be differentiated in the order. Substitute $s = N + \varepsilon$ into (6.8), expand $\sin(\pi\varepsilon)$, and differentiate. The leading term in (6.19) is $-2\pi i(-1)^N r E_{N,r-1}$, whose coefficient is nonzero. This proves triangular invertibility and the generating identity. □

For example,

$$E_{N,1} = \frac{(-1)^{N+1}}{4\pi i} D_{N,2}, \qquad E_{N,2} = \frac{(-1)^{N+1}}{6\pi i} D_{N,3} + \frac{\pi^2}{3} E_{N,0}.$$

The word "recovery" has a precise scope: the complete analytic family of order jets, or its indicated discontinuities, is supplied. The terminating polynomial B_N alone contains no such information.

6.3 Exact large-order coefficients and polygamma jets

Proposition 6.5 (The coefficient tail after differentiation)**.** *For $2k > N \geq 0$,*

$$\boxed{a'_{2k}(N) = 2(-1)^{N+1}\eta_{\mathrm{D}}(2k)N!(2k-N-1)!.} \tag{6.22}$$

More generally, the exact analytic germ

$$\frac{a_{2k}(N+\varepsilon)}{\varepsilon a'_{2k}(N)} = \frac{\sin(\pi\varepsilon)}{\pi\varepsilon}\frac{\Gamma(N+1+\varepsilon)}{\Gamma(N+1)}\frac{\Gamma(2k-N-\varepsilon)}{\Gamma(2k-N)} \tag{6.23}$$

determines all higher jets. In particular, if $\Delta = \psi(N+1) - \psi(2k-N)$, then

$$a''_{2k}(N) = 2a'_{2k}(N)\Delta, \tag{6.24}$$

$$a'''_{2k}(N) = 3a'_{2k}(N)\left\{\Delta^2 + \psi_1(N+1) + \psi_1(2k-N) - \frac{\pi^2}{3}\right\}. \tag{6.25}$$

Proof. Reflection of the reciprocal gamma factor gives, away from removable specializations,

$$a_{2k}(s) = -\frac{2\eta_{\mathrm{D}}(2k)}{\pi}\Gamma(s+1)\sin(\pi s)\Gamma(2k-s).$$

For $2k > N$, both gamma factors on the right are regular at $s = N$. Differentiating the simple zero of the sine proves (6.22). Dividing by its linear term gives (6.23), and logarithmic differentiation gives the displayed higher derivatives. □

For a compact all-orders coefficient prescription, let Y_j be the complete exponential Bell polynomials, characterized by

$$\exp\left(\sum_{j\geq 1} L_j\frac{u^j}{j!}\right) = \sum_{j\geq 0} Y_j(L_1,\ldots,L_j)\frac{u^j}{j!}.$$

Then, for $r \geq 1$ and $2k > N$,

$$a^{(r)}_{2k}(N) = r\,a'_{2k}(N)Y_{r-1}(L_1,\ldots,L_{r-1}), \tag{6.26}$$

where

$$L_j = \psi^{(j-1)}(N+1) + (-1)^j\psi^{(j-1)}(2k-N) - 2\mathbf{1}_{\{j \text{ even}\}}(j-1)!\zeta(j). \tag{6.27}$$

The final term is omitted for odd j, so $\zeta(1)$ is never evaluated. This makes the connection between order jets, polygamma values, harmonic numbers, and factorial divergence explicit.

6.4 A meromorphic Borel kernel for the full first-order jet

The ordinary transform in Theorem 6.1 has logarithmic singularities after differentiating at an integer. A different but compatible normalization gives only simple poles. This simplification requires restoring $x^s/\Gamma(s+1)$ before taking the order derivative.

Define

$$L_N(x) = \sum_{0\leq 2k\leq N} c_{2k}\frac{x^{N-2k}}{(N-2k)!}\{\log x - \psi(N+1-2k)\}, \tag{6.28}$$

$$C_N(p) = \sum_{0\leq 2k\leq N} c_{2k}p^{2k}, \tag{6.29}$$

$$K_N(p) = (-1)^{N+1}p^{-N-1}\left\{\frac{\pi p}{\sin(\pi p)} - C_N(p)\right\}. \tag{6.30}$$

The singularity of K_N at 0 is removable. The generating identity

$$\frac{\pi p}{\sin(\pi p)} = 1 + 2\sum_{k\geq 1} \eta_{\mathrm{D}}(2k) p^{2k}, \qquad |p| < 1, \tag{6.31}$$

shows that K_N is the minor Borel transform of the negative-power tail in the full first-order jet:

$$\partial_s \frac{x^s A_s(x)}{\Gamma(s+1)}\Big|_{s=N} = L_N(x) + 2(-1)^{N+1} \sum_{2k>N} \eta_{\mathrm{D}}(2k)(2k-N-1)! x^{N-2k}. \tag{6.32}$$

Here "minor" means $x^{-r} \mapsto p^{r-1}/\Gamma(r)$ for $r \geq 1$; its inverse is the Laplace integral without an extra factor of x.

For K_N, lateral integration has the ordinary contour meaning of passing above or below every positive integer. Equivalently, it is

$$\int_0^{\infty,\pm} e^{-xp} K_N(p)dp = \mathrm{PV}\int_0^{\infty} e^{-xp} K_N(p)dp \mp i\pi \sum_{m\geq 1} e^{-mx} \operatorname{Res}_{p=m} K_N(p). \tag{6.33}$$

The principal values are symmetric at each pole; their sum converges after exponential weighting. These pole integrals are not claimed to be ordinary absolutely convergent real integrals.

Theorem 6.6 (Exact meromorphic representation of the first jet)**.** *For every integer $N \geq 0$ and every real $x > 0$,*

$$\boxed{\partial_s G_s^{\pm}(x)\big|_{s=N} = L_N(x) + \int_0^{\infty,\pm} e^{-xp} K_N(p)dp.} \tag{6.34}$$

Its poles encode the full exponentially small Fermi–Dirac series:

$$\operatorname{Res}_{p=m} K_N(p) = \frac{(-1)^{N+m+1}}{m^N}, \qquad m \geq 1. \tag{6.35}$$

Taking the symmetric average yields the real formula

$$\partial_s F_s(x)\big|_{s=N} = L_N(x) + \mathrm{PV}\int_0^{\infty} e^{-xp} K_N(p)dp - (-1)^N E_{N,1}(x). \tag{6.36}$$

Proof. At $N = 0$, differentiating (6.4) gives, initially for $|p| < 1$,

$$\begin{aligned} \partial_s B_s(p)\big|_{s=0} &= \sum_{m\geq 1} (-1)^{m-1} \log(1 - p^2/m^2) \\ &= \log\left\{ \frac{\pi p}{2} \cot \frac{\pi p}{2} \right\}. \end{aligned} \tag{6.37}$$

The last equality follows from the Euler product for sine, and the logarithm denotes the analytic germ vanishing at zero, continued on the same lateral sheet. It is not an instruction to take the principal logarithm separately at every positive real p. Its derivative is

$$\frac{1}{p} - \frac{\pi}{\sin(\pi p)} = K_0(p).$$

Differentiate the prefactor in (6.6) and integrate by parts on an upper or lower ray. Boundary terms vanish at zero and infinity, giving (6.34) for $N = 0$, since $L_0 = \log x + \gamma$. Taking the boundary limit of the ray gives precisely (6.33). This also proves convergence of these lateral and principal-value integrals. For general N, the subtracted kernel has at most polynomial growth along such rays and is regular at zero, so the same convergence statements hold directly.

To pass from $N-1$ to N, note first that

$$\frac{d}{dx}G_s^{\pm}(x) = G_{s-1}^{\pm}(x),$$

by (6.7) and $\frac{d}{dx}F_s(x) = F_{s-1}(x)$. Therefore the left side of (6.34), differentiated in x, is its counterpart with $N-1$. On the right side,

$$L_N'(x) = L_{N-1}(x) + \mathbf{1}_{\{N \text{ even}\}}\frac{c_N}{x},$$
$$-pK_N(p) = K_{N-1}(p) - \mathbf{1}_{\{N \text{ even}\}}c_N.$$

Thus the extra c_N/x terms cancel after Laplace integration. By induction the difference of the two sides is constant in x. Both sides minus $L_N(x)$ tend to zero as $x \to +\infty$. Here is the order-uniform detail on the left. Fix a circle $|s-N| = a < 1/2$, choose K with $2K \geq N$, and truncate the full normalized expansion after $k = K$. Its uniform remainder on this circle is $O(x^{N+a-2K-2})$. Cauchy's formula bounds its order derivative at N by $O(a^{-1}x^{N+a-2K-2}) = o(1)$. The derivatives of the finitely many terms with $2k > N$ are negative powers of x and also tend to zero; the remaining derivatives are exactly $L_N(x)$. On the right, the lateral pole integral can be deformed to any fixed upper or lower ray of angle less than $\pi/2$. On that ray its kernel is regular at zero and grows at most polynomially at infinity. Elementary Laplace estimates therefore show that the integral tends to zero. The constant vanishes. The residue calculation follows from $\sin(\pi(m+t)) = (-1)^m\pi t + O(t^3)$. Finally, differentiation of (6.7) at $s = N$ gives

$$\partial_s G_s^{\pm}(x)\big|_{s=N} = \partial_s F_s(x)\big|_{s=N} + (-1)^N E_{N,1}(x) \mp i\pi(-1)^N E_{N,0}(x),$$

whose symmetric average proves (6.36). □

For example,

$$K_0(p) = \frac{1}{p} - \frac{\pi}{\sin(\pi p)}, \qquad K_1(p) = \frac{\pi p/\sin(\pi p) - 1}{p^2}.$$

The physical first-order correction $E_{N,1}$ starts at e^{-2x}, because $\log 1 = 0$, whereas the Stokes difference of the same first jet already starts at e^{-x}. These are different exact statements and should not be conflated.

6.5 A convergent inverse expansion at integer order

The exponential completion also gives a direct nonlinear application. The following formula is a specialization of analytic Lagrange inversion, with every exponential sector given explicitly.

Proposition 6.7 (All exponential sectors of the integer-order inverse)**.** *Fix an integer $N \geq 1$. Let X be sufficiently large and real, put $y = P_N(X)$ and $V(X) = P_N'(X)$, and let x be the real solution of $F_N(x) = y$. Define, for $1 \leq r \leq m$,*

$$C_{m,r}^{(N)} = \sum_{\substack{m_1+\cdots+m_r=m \\ m_j \geq 1}} \frac{1}{(m_1\cdots m_r)^N}, \qquad \mathcal{D}_m = \frac{1}{V(X)}\left(\frac{d}{dX} - m\right).$$

Then the expansion

$$x = X + \sum_{m\geq 1} e^{-mX}R_{N,m}(X) \tag{6.38}$$

converges, and all its coefficients lie in $\mathbb{Q}(\pi^2)(X)$, with the explicit rational-function formula

$$R_{N,m}(X) = \sum_{r=1}^{m} \frac{(-1)^{Nr+m-r}}{r!} C_{m,r}^{(N)} \mathcal{D}_m^{r-1}\left(\frac{1}{V(X)}\right). \tag{6.39}$$

In particular, with $\sigma = (-1)^N$,

$$x = X + \frac{\sigma}{V}e^{-X} - \left\{\frac{\sigma}{2^N V} + \frac{1}{V^2} + \frac{V'}{2V^3}\right\} e^{-2X} + O(e^{-3X}) \tag{6.40}$$

as $X \to +\infty$. *The implied constant can be chosen uniformly once* N *is fixed and* X *is large enough.*

Proof. For a complex auxiliary variable z, consider the implicit equation

$$P_N(X+\delta) - P_N(X) = \sigma \sum_{j\geq 1} \frac{(-1)^{j-1}}{j^N} z^j e^{-j\delta}. \tag{6.41}$$

It has a unique analytic solution $\delta = \delta_X(z)$ at $z = 0$, since $V(X) \neq 0$. It is analytic in a disk $|z| < r_0$ with $r_0 > 0$ independent of sufficiently large X. For completeness, take $r_0 = 1/(4e)$. On $|\delta| = 1$ the right side has modulus at most $1/3$. The left side has modulus asymptotic to $V(X)$ uniformly on that circle, and has only the simple zero $\delta = 0$ inside it for large X. For $N = 1$ the left side is exactly δ; for $N \geq 2$, $V(X) \to \infty$. Rouché's theorem gives exactly one simple zero within $|\delta| < 1$, throughout $|z| < r_0$, and hence the stated analytic solution and a uniform bound $|\delta_X(z)| < 1$.

For the coefficient calculation, keep the base X fixed, let g be the local inverse of P_N, and set

$$h_{X,z}(w) = \sum_{j\geq 1} \frac{(-1)^{j-1}}{j^N} z^j e^{-j(w-X)}.$$

Introduce a scalar t multiplying the right side of (6.41). Lagrange–Bürmann inversion gives

$$\delta(t) = \sum_{r\geq 1} \frac{(t\sigma)^r}{r!} \left(\frac{d}{dv}\right)^{r-1} \{g'(v) h_{X,z}(g(v))^r\}\Big|_{v=y}.$$

Now

$$h_{X,z}(g(v))^r = \sum_{m\geq r} (-1)^{m-r} C_{m,r}^{(N)} z^m e^{-m(g(v)-X)}.$$

Since $d/dv = V(g(v))^{-1} d/dg(v)$ and $g(y) = X$, differentiating a term of this sum replaces differentiation by $\mathcal{D}_m = V(X)^{-1}(d/dX - m)$ on its coefficient. Only $r \leq m$ contributes to z^m, so setting $t = 1$ coefficientwise gives (6.39) without requiring an independent convergence assertion in t. By uniqueness of the formal implicit solution, these are the Taylor coefficients of the already established analytic function $\delta_X(z)$. Substituting $z = e^{-X} < r_0$ proves convergence and the physical inverse equation, using (6.16). The physical real inverse is unique because $F_N'(x) = F_{N-1}(x) > 0$ for real x and $N \geq 1$. The coefficient formula with $m = 1, 2$ gives (6.40); a Cauchy estimate on a fixed smaller disk gives the uniform remainder bound. □

For $N = 1$, $P_1(X) = X$ and the inverse is $x = \log(e^y - 1) = y - \sum_{m\geq 1} e^{-my}/m$, providing an exact elementary check. For $N = 2$,

$$X = \sqrt{2y - \pi^2/3}, \qquad x = X + \frac{e^{-X}}{X} - \left(\frac{1}{4X} + \frac{1}{X^2} + \frac{1}{2X^3}\right) e^{-2X} + O(e^{-3X}).$$

6.6 Interpretation of the Borel identities

The principal points requiring care are the shift of Fermi–Dirac order, the factor $x^s/\Gamma(s+1)$ before order differentiation, the reversed sign of the boundary argument of $1 - p/m$, and the distinction between an integrable branch point and a pole principal value. The paired summands in (6.4) justify the stronger domain $\operatorname{Re} s > -1$; integrating the derivative B_s' directly would lose that domain unless regularized. At $N = 0$ the logarithm in (6.37) must be continued from its germ, because a pointwise principal logarithm would discard accumulated jumps.

7 What the two asymptotic regimes have in common

The exact formulas exhibit a specific common mechanism. Near $t = 0$, the Gamma pole and the Hurwitz-zeta pole cancel, leaving the entire function of the scaled order

$$\frac{1 - e^{-X\varepsilon} E_p(\varepsilon)}{\varepsilon}.$$

At infinity, the reciprocal-Gamma zero is expressed by the sine factor in (6.23). Specializing the order before examining its derivatives removes information in both cases. Keeping the order as an analytic variable makes that information explicit and allows a finite coefficient calculation.

There is also a useful distinction. The local Taylor tails in section 3 are convergent, and the inverse uses a holomorphic function of small Lambert coordinates. The Sommerfeld tail after differentiation is factorially divergent; the exact lateral integrals supply its analytic meaning. Calling both expansions transseries does not replace either of these separate convergence arguments.

7.1 A second order–logarithm crossover, now in coefficient index

The exact coefficient germ has an immediate consequence that connects the local resonance scale with the large-order scale. It also identifies a natural starting point for a uniform optimal-truncation problem.

Proposition 7.1 (Crossover of the revived factorial coefficients)**.** *Fix $N \geq 0$, let $2k > N$, and put $\varepsilon = \lambda/\log(2k)$. Then, uniformly for λ in any fixed compact subset of $\mathbb{C}$,*

$$\frac{a_{2k}(N+\varepsilon)}{\varepsilon a'_{2k}(N)} = e^{-\lambda}\left\{1 + \frac{\psi(N+1)\lambda}{\log(2k)} + O\left(\frac{1}{\log^2(2k)} + \frac{1}{k}\right)\right\} \qquad (k \to \infty). \tag{7.1}$$

The quotient at $\lambda = 0$ is interpreted by its removable value.

Proof. In (6.23), the sine quotient is $1+O(\varepsilon^2)$, while $\Gamma(N+1+\varepsilon)/\Gamma(N+1) = 1+\psi(N+1)\varepsilon+O(\varepsilon^2)$. The gamma-ratio asymptotic, uniformly for ε in a small fixed disk, gives

$$\frac{\Gamma(2k-N-\varepsilon)}{\Gamma(2k-N)} = (2k-N)^{-\varepsilon}\{1 + O(k^{-1})\}.$$

This follows, for example, by subtracting the uniform Stirling expansions of the two logarithms of Gamma [7]. Since $\varepsilon\log(2k-N) = \lambda+O(k^{-1}/\log(2k))$ on the specified compact set, multiplying these estimates proves the formula. □

Thus the order displacement is measured against $\log(1/t)$ in the local problem and against $\log k$ in the coefficient problem. The leading profiles are different because one construction pairs poles and the other resolves a zero. The common variable is the product of a small order displacement and a large logarithm.

7.2 A precise meaning of reconstruction

The finite transformation (3.16) recovers a pure logarithmic basis from a finite list of order jets. The triangular relation (6.19) recovers exponential coefficients from the discontinuities of another list of order jets. Both are exact, invertible triangular operations once their input families and normalizations are specified.

Neither statement says that one finite terminating expansion determines all exponentially small terms of an arbitrary function. One may add a flat term to a function without altering its Poincaré expansion. The recovery works here because the analytic polylogarithm family, its Borel

transform, and its continuation have all been supplied. Likewise, the logarithmic monodromy in (3.22) acts on the argument plane, while (6.8) compares two summation directions in a Borel plane. Their exact formulas are related through the order parameter, but they are not the same operator.

The inverse theorems add a nonlinear step to this picture. At the endpoint, arithmetic cancellation of tail coefficients removes terms from the support of a convergent Lambert chart. At infinity, the residue-generated exponential series feeds into the finite composition sums of (6.39). Near a complex critical value, inversion changes the scale itself because the limiting root has multiplicity two. This explains why the article requires three analytic arguments: an ordinary implicit function theorem, a quadratic blowup, and a Borel–Laplace identity.

8 Audit, verification, and integration status

8.1 What was audited and what should be corrected

The repository review was targeted at the predecessor reports needed for the theorems above, their definitions and proofs, the surrounding research questions, and the relevant programme documentation. It was not a line-by-line audit of every preserved report in either directory. The package records the inspected commit and exact paths in its integration notes.

No new mathematical error was confirmed in the directly used repository resonance theorem, the single-logarithm Lambert theorem, or the confluent real-exponent inverse chart [1, 2, 3]. The natural repository change is an addition and cross-reference: theorem 2.1 resolves the fixed-sheet closed-sector form of question R1, while sections 3 to 6 develop its inverse and order-derivative consequences. The real-axis predecessor should retain its existing hypotheses and attribution.

A concrete correction is needed for the external principal-sheet accumulation claim discussed after proposition 5.5. The correct modulus is $|q^{-s}| = |q|^{-\Re s} e^{(\Im s)\arg q}$ on a fixed logarithm branch. At any fixed order with real part $1/2$, the singular leading term therefore prevents an infinite sequence of principal-sheet zeros from approaching $z = 1$. This statement is about a fixed order; it does not rule out many nearby finite zeros or limits in which the order also varies.

A second external scope correction concerns O'Sullivan's overview in §8.1 of *Zeros of the dilogarithm* [19]. Its blanket finiteness assertion for principal-branch zeros when $\Re s > 0$ cannot hold on the whole line $\Re s = 1$: corollary 5.3 and proposition 5.4 construct infinitely many zeros of $\mathrm{Li}_{1+i\beta}$ for each sufficiently small $\beta \neq 0$. They lie inside $|z| < 1$, since $\Re t_{c,k} > 0$. The author-hosted text is linked as the published version on the author's research page. The hypotheses of the original Gawronski paper cited there [20] were not available for inspection; the correction concerns the overview as written and is not an assessment of that original theorem.

Two errors were caught and removed during preparation of this report. First, an asymptotically linear inverse was initially described as analytic: the logarithmic term at the endpoint makes that description false. The corrected statement appears before (3.3). Second, the first draft of the inverse calculation at infinity omitted an auxiliary multiplier from a displayed Lagrange series and blurred the distinction between its base point and its independent exponential variable. The proof of proposition 6.7 now keeps both variables explicit. Neither issue is attributed to the repository.

A further set of scope distinctions is mathematical, not editorial. The pole $\eta = p/q$ in theorem 4.3 is a pole of auxiliary coefficient functions, not an established nearest singularity of the physical inverse. Exhaustion in proposition 5.4 concerns a small neighborhood and a sufficiently small fixed imaginary order. A sequence of local branch values does not by itself prove a natural boundary for a specified global inverse. Finally, the ordinary lateral integrals in theorem 6.1 and the pole principal values in theorem 6.6 have different meanings and are not interchangeable definitions.

8.2 Symbolic checks

The package includes four verification programs and their complete machine-readable output. The symbolic checks use exact arithmetic and independent finite recurrences where possible.

Identity checked	Cases
Finite Gamma conjugation, $p = 1, 2, 3$, $q = 1, \ldots, 5$	15
Implicit inverse recurrence versus the Lagrange formula	60
Sharp coefficient-pole constants	60
Explicit v_1, v_2 and second-derivative formulas	36
Revived factorial coefficients at integer order	47
Cosecant generating coefficients	9
Residues of the meromorphic first-jet kernel	30
Triangular discontinuity identities	7
Integer-order inverse composition coefficients	6
Total finite symbolic cases	270

All recorded cases passed. These checks catch sign, factorial, normalization, and coefficient errors. They do not replace the normal-convergence, implicit-function, branch, or uniform-remainder arguments in the proofs.

8.3 Independent numerical evaluation routes

For local normalized jets, actual logarithmically weighted defining series were compared with the convergent analytic-tail formula at four complex values of t. The cases include $a = 1$ and $a = 1/2$, $p = 1, 2, 3$, and $q = 1, 2, 3$. With 85-digit arithmetic and explicit exponential bounds for the omitted defining-series terms, the absolute discrepancies ranged from 1.2×10^{-44} to 1.8×10^{-54}.

Two inverse examples use the same defining-series evaluator for the final residual, rather than evaluating only the model being inverted. At the common Lambert core $T = 0.01$, the errors in successive approximations to t were:

(p, q, a)	Core T	Through v_1	Through v_2
$(1, 2, 1)$	$9.44 \cdot 10^{-6}$	$1.49 \cdot 10^{-8}$	$3.01 \cdot 10^{-11}$
$(2, 3, 1/2)$	$1.63 \cdot 10^{-6}$	$7.61 \cdot 10^{-10}$	$1.64 \cdot 10^{-13}$

The full numerical roots gave defining-series residuals approximately 1.1×10^{-74} and 1.0×10^{-77}.

At infinity, the symmetric first-jet Borel expression $L_N + \mathrm{PV} \int e^{-x\xi} K_N(\xi)\, d\xi$ was compared with the symmetric logistic-density order derivative of G_s at $s = N$. This reference includes the reflected order-derivative contribution in (6.36); it is not simply the physical derivative of $F_s(x)$. At $x = 4$, 60-digit arithmetic and a principal-value integration cutoff after 30 positive poles gave:

N	0	1	2	3
Absolute discrepancy	$8.17 \cdot 10^{-54}$	$2.67 \cdot 10^{-55}$	$1.30 \cdot 10^{-55}$	$4.23 \cdot 10^{-57}$

The complex-resonance program uses 220-digit arithmetic. It checks nine critical-lattice samples, eighteen split-root samples, and twelve unfolded-root samples. The sampled critical orders are $\beta = 0.1, 0.2, -0.15$ with $|k| = 1, 2, 3$; the split tests include $k = 1, -1, 2$ and $Y = 80, 160, 320$. The largest direct critical-point residual is 2.65×10^{-140}, and the largest relative difference between direct and paired critical values is 1.46×10^{-139}. The direct original

defect equation for the eighteen split roots has relative residual at most 1.47×10^{-82}. The paired split and unfolding equations have residuals below 1.8×10^{-220} and 1.2×10^{-218}, respectively.

A fourth program checks the branch convention by an independent Bose integral. For $s = 1 + 0.2i$ it evaluates the root

$$t \approx 2.2677571996479048744 \cdot 10^{-14} - 4.1525596683989631880 \cdot 10^{-15} i$$

using

$$\mathrm{Li}_s(e^{-t}) = \frac{1}{\Gamma(s)} \int_{-\infty}^{\infty} \frac{e^{sv}}{e^{e^v + t} - 1}\, dv.$$

This is the standard integral representation after the real substitution $u = e^v$. In particular, $\Re t > 0$ directly gives $|e^{-t}| < 1$. At 90 digits, quadrature over $[-300, 8]$ gives a residual 2.03×10^{-90}. The omitted integration tails have analytic bounds 2.35×10^{-117} on the left and less than 10^{-1294} on the right. These cutoff bounds do not certify quadrature rounding. The separate integral test addresses the branch question independently of the local formula used to locate the root.

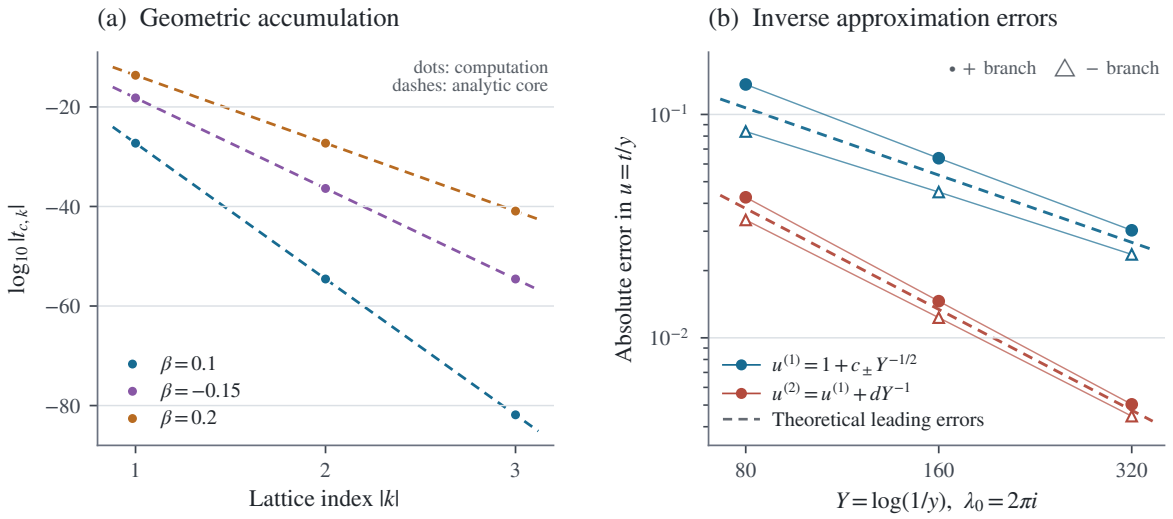


Figure 1: Complex resonance and inverse branching. (a) Recorded critical points for $\varepsilon = i\beta$, $\beta \in \{0.1, -0.15, 0.2\}$, $|k| = 1, 2, 3$. The dashed lines are $\log_{10} |a(i\beta, 0)| - 2\pi|k|/(|\beta| \log 10)$; the discrepancies are too small to resolve at this scale. (b) Absolute errors in $u_\pm = t_\pm / y$ for $\varepsilon = 2\pi i / Y$, $Y = 80, 160, 320$. Circles and triangles distinguish the two branches. First and second approximations retain terms through $Y^{-1/2}$ and Y^{-1}, respectively. The reference curves $|d|Y^{-1}$ and $|e_\pm|Y^{-3/2}$ use the exact coefficients derived in appendix A; none is fitted to the data.

All these numerical figures are diagnostics, not interval certificates. The small residual of an equation near a critical point is not by itself a rigorous root-error bound. Furthermore, direct subtraction of the two order-one terms in the endpoint defect loses digits when $y = e^{-Y}$ is extremely small; the paired normal form avoids that subtraction. Preliminary differentiation of a generic numerical polylogarithm routine precisely at integer order was also less reliable than the defining-series and logistic-integral routes. The verification programs therefore keep those routes explicit.

8.4 Reproducibility and repository integration

The source package contains one complete article source, its PDF, the vector figure and plotting program, the four verification programs, their JSON records, a dependency file, a build command, and a manifest of checksums. All mathematical arguments are in the article; running Python is not necessary to follow a proof. Running the programs is useful for reproducing the finite diagnostics.

The integration notes propose one canonical article location and cross-links from both programmes. They preserve the predecessor reports and distinguish a theorem extension from an erratum. No repository file was edited as part of this report. The supplied package is intended for review and subsequent integration.

9 Further research questions and a proposed programme

The questions below are not assertions of results established here. They are selected to use the exact formulas while addressing limitations that remain after the local existence and coefficient calculations have been completed.

9.1 Certified domains and the geometry of inverse coefficients

Q1. Obtain effective convergence domains. For actual tails $R_{p,q,a}$, determine explicit polydiscs and computable constants for (4.13), the critical charts a, b, and the split charts $W_{\pm}$. A practical starting point is the factorial majorant in theorem 2.1, followed by an explicit contraction or interval Newton estimate for the implicit equation. A useful deliverable would accept p, q, a, a target precision, and a sector, and return both an approximation and a certified error. The main difficulty is retaining useful bounds near a fold, where an ordinary inverse condition number diverges and the quadratic coordinate must be used instead.

Q2. Determine large coefficient-index asymptotics. The exact pole order of $v_n(\eta)$ as a function of η is known from theorem 4.3. What is the asymptotic behavior of $v_n(\eta)$ as $n \to \infty$ at a fixed admissible η? For a given polylogarithm tail, locate the dominant solutions of the coupled equations

$$H(T, \eta, V) = 0, \qquad H_V(T, \eta, V) = 0,$$

where H is the left side of (4.4) minus 1. Decide when a simple quadratic singularity supplies an $n^{-3/2}$ coefficient factor, which competing singularities have the same modulus, and how the answer changes as η approaches p/q. This would connect the Catalan pole law with actual convergence radii, without identifying those two different questions prematurely.

Q3. Classify arithmetic cancellations and missing sectors. For finitely many shifts a_ν and weights w_ν, consider $\sum_\nu w_\nu \mathcal{Y}_{p,q,a_\nu}$. The first h analytic tail coefficients vanish precisely when the explicit zeta-derivative expressions in (3.14) satisfy h linear relations. Classify cancellations that persist for infinitely many coefficient indices, especially for rational shifts or roots-of-unity combinations. The half-shift parity at $q = 1$ is an elementary example; differentiation in the order generally removes those zeros. Determine which distribution identities preserve a nontrivial support restriction for $q \geq 2$, and what restrictions remain after compositional inversion.

9.2 Complex-order critical values and global inverse structure

Q4. Extend the critical classification beyond small imaginary order. The local critical lattice is now exhausted for sufficiently small fixed $\beta \neq 0$. For larger β, the constant-term

balance suggests the limiting argument

$$\theta_\beta = -\frac{1}{\beta}\log\left|\frac{i\beta\,\zeta(1+i\beta)}{\Gamma(1-i\beta)}\right|.$$

Determine a complete principal-sheet criterion in terms of this quantity and the analytic tail, including the boundary cases $\theta_\beta = \pm\pi$. Establish how the lattice enters or leaves a chosen logarithm sheet, and quantify its critical-value asymptotics uniformly in the remaining parameter ranges. A statement on the logarithmic cover must be distinguished from one on a fixed principal sheet.

Q5. Describe which inverse sheets meet at the accumulating values. The local square-root branch values of $F_{i\beta}^{-1}$ accumulate at zero. Which of them are accessible from a particular inverse germ by continuation along specified paths? What is the monodromy action on that connected family of inverse sheets? Can a description of the complete local lattice be converted into precise boundary behavior for an inverse branch on a slit or spiraling domain? Accumulation of branch values alone does not answer whether a chosen inverse has a natural boundary, and that global inference remains to be proved or disproved.

Q6. Classify higher-multiplicity inverse resonances. Replace the dilogarithm defect by a higher Taylor-subtracted remainder, a normalized order jet, or a finite linear combination of such jets. Determine which parameter choices make the limiting inverse equation have multiplicity $m > 2$. A proved classification should identify the minimal unfolding parameters, the analogue of the exact branch curve in (5.42), and the fractional inverse-logarithmic scale forced by the multiplicity. The natural procedure is to compute the limiting logarithmic polynomial exactly before applying a Weierstrass or Newton-polygon analysis. It is not enough to infer an mth-root scale from a numerical collision of approximate roots.

9.3 Uniform resurgence and nonlinear reconstruction

Q7. Match the revived coefficient crossover with optimal truncation. Proposition 7.1 already resolves the coefficient-level double limit with $\varepsilon \log k$ bounded. Develop a function-level expansion uniform when $x \to \infty$, $s \to N$, and $\varepsilon \log x$ remains bounded. It should interpolate between the terminating integer-order polynomial, the factorially divergent order jet, and the exact exponential completion. The next step is to prove an explicit remainder for a truncation index proportional to x, and then carry that estimate through the nonlinear inverse. The exact gamma-product formula and the paired Borel integral give two independent starting points for this analysis.

Q8. Derive Lerch and cyclotomic analogues of the residue kernel. For periodically weighted Dirichlet series, identify the analogue of $\pi\xi/\sin(\pi\xi)$ in (6.30). Determine its poles, the rays on which they lie, the coefficients recovered by the first order jet, and the effect of shift differentiation. For root-of-unity weights one expects finite combinations of elementary trigonometric kernels, but the precise branch factors and the correct physical completion must be derived. The goal is an exact formula with a specified lateral convention and domain, followed by a nonlinear inverse application.

Q9. Compute noninteger inverse Stokes transformations. At integer order (6.39) gives every exponential inverse sector through finite composition sums. For noninteger order, determine the discontinuity of a specified sectorial compositional inverse from the exact discontinuity (6.8). Relate its first derivative in the order to the integer residue formula, while controlling the deformation of the algebraic core. This would test how much of the triangular reconstruction survives a nonlinear change of dependent variable.

Q10. Establish uniform complex-argument Borel formulas. The exact lateral identities here are proved for real $x > 0$. Continue them to complex sectors with explicit compatible polylogarithm branches and quantitative bounds as a summation ray approaches a pole direction. Determine the connection formulas when the Laplace contour must rotate across several rays. The integral representation suggests the extension, but the contour deformation, the argument of x^{s+1}, and the reflected polylogarithm must be tracked together.

9.4 Recommended order of work

For repository development, the most direct next project is Q1: it turns the convergent charts into certified evaluators and supplies sharper test cases for the other questions. Q2 and Q4 then address the actual singularity geometry. Q7 and Q9 are the most substantial asymptotic continuation of the present order-jet idea. Q3 and Q8 provide the strongest connection to the arithmetic side of the polylogarithm programme.

Before a publication-priority claim, carry out a dedicated comparison with the full original literature on complex polylogarithm zeros, near-integer order evaluation, and complete Fermi–Dirac expansions. The proof status of the formulas in this report and the priority status of their presentation are separate matters.

A The coefficient behind the numerical error reference

Write $\lambda = 2\pi i$, $r_1 = \gamma - 1$, and

$$r_2 = \frac{r_1^2 + \zeta(2) + 1}{2}, \qquad r_3 = \frac{r_1^3}{6} + \frac{r_1(\zeta(2)+1)}{2} + \frac{\zeta(3)-1}{3}, \qquad D = r_2 + \gamma_1, \qquad J = \frac{\gamma_2}{2} - r_3.$$

Here $\zeta(1+w) = w^{-1} + \gamma - \gamma_1 w + \gamma_2 w^2/2 + \cdots$. For the two local branches, choose $c_\pm = \pm\sqrt{-2\lambda D}$, with the principal square root for c_+. The next coefficient after $d = -\lambda(r_1 + 4D/3)$ is

$$e_\pm = \frac{\lambda^2}{c_\pm}\left(J + \frac{r_1^2}{2} + (3\gamma - 2)D + \frac{13D^2}{9}\right).$$

Thus

$$\frac{t_\pm}{y} = 1 + \frac{c_\pm}{\sqrt{Y}} + \frac{d}{Y} + \frac{e_\pm}{Y^{3/2}} + O(Y^{-2}).$$

To verify the formula directly, set $x = Y^{-1}$, $u = 1 + w$ and expand the exact paired equation, omitting its exponentially small analytic remainder:

$$\begin{aligned} 0 = & -\frac{w^2}{2} + \frac{w^3}{6} - \frac{w^4}{12} - \lambda x\left(D + (D + r_1)w + \frac{\gamma w^2}{2}\right) + \lambda^2 J x^2 \\ & + O(w^5 + xw^3 + x^2 w + x^3) + O(e^{-Y}). \end{aligned}$$

The coefficients of x, $x^{3/2}$ and x^2 give $c_\pm$, d and $e_\pm$ in that order. The analytic square-root chart justifies this coefficient calculation and its remainder. The reference curves in panel (b) are therefore $|d|/Y$ and $|e_\pm|/Y^{3/2}$, where

$$|d| \simeq 8.561352970598396, \qquad |e_+| = |e_-| \simeq 27.13447947885805.$$

B Literature-search boundaries

The primary mathematical inputs used for the proofs were inspected in full at the relevant formulas or sections. They include the DLMF local expansions, the predecessor repository proofs,

the near-integer computational formulas, and the full author manuscript of Gil–Segura–Temme. The bibliography supplies stable locations rather than relying on search-result snippets for those inputs.

There is classical precedent for oscillatory boundary accumulation. Gawronski and Stadtmüller [21] exhibit infinitely many zeros in $(0,1)$ for the series $\sum_{n\geq 0}\sin(\log(n+1))z^n$. This is a combination of complex-order polylogarithms rather than the single-polylogarithm family studied here. It is a reason to regard the explicit common charts, critical values, and inverse unfolding as the relevant increments, rather than claiming a new general mechanism for boundary accumulation.

The search also identified sources requiring fuller comparison before a claim of publication priority. Fornberg–Kölbig's *Complex zeros of the Jonquière or polylogarithm function* [22] and Paulsen's *The Behavior of Polylogarithms at $z = 1$* [23] were located through primary bibliographic records and available descriptions, but their full texts were not obtained in this review. Gawronski's 1979 paper [20] likewise requires inspection of the original hypotheses. These access limits do not enter any proof in the article. They do limit what the review can say about the historical priority of the local zero-lattice mechanism.

The most appropriate present description is therefore a proved research report containing explicit extensions, a coherent order-jet framework, and precisely scoped corrections. The contribution claims refer to the formulas and arguments actually established here.

References

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